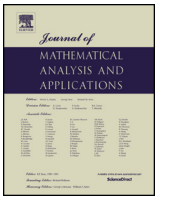
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Explicit bounds for the Riemann zeta function and a new zero-free region



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ABSTRACT

We prove that $|\zeta(\sigma+it)| \leq 70.7|t|^{4.438(1-\sigma)^{3/2}} \log^{2/3}|t|$ for $1/2 \leq \sigma \leq 1$ and $|t| \geq 3$. As a consequence, we improve the explicit zero-free region for $\zeta(s)$, showing that $\zeta(\sigma+it)$ has no zeros in the region $\sigma \geq 1 - 1/\left(53.989(\log|t|)^{2/3}(\log\log|t|)^{1/3}\right)$ for $|t| \geq 3$ and asymptotically in the region $\sigma \geq 1 - 1/\left(48.0718(\log|t|)^{2/3}(\log\log|t|)^{1/3}\right)$ for $|t|$ sufficiently large.

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1. Introduction

Let $\zeta(s)$ be the Riemann zeta function, with $s = \sigma + it$ a complex variable. It is known that all the non trivial zeros of $\zeta(s)$ have real part $\sigma \in (0,1)$. Detecting zero free regions for $\zeta(s)$ inside the critical strip $0 < \sigma < 1$ is an open problem that has stimulated much interest in analytic number theory. Great effort has been put in trying to find both asymptotically and explicit regions inside the critical strip where there are no zeros of $\zeta(s)$. The classical zero-free region is of the form $\sigma > 1 - 1/(R_0 \log|t|)$, where R_0 is a positive constant. The best known result of this form is due to Mossinghoff, Trudgian and Yang [18] with $R_0 = 5.558691$ for every $|t| \geq 2$ (see [23,22,13,12,10,17] for previous results). The Littlewood zero-free region [16] is instead of the form of $\sigma > 1 - \log\log|t|/C_1 \log|t|$, where C_1 is a positive constant. It has been made first explicit by Yang [32], who found $C_1 = 21.233$ for $|t| \geq 3$.

Asymptotically larger zero-free regions for the Riemann zeta function $\zeta(s)$, known as Korobov–Vinogradov zero-free regions, are of the form

$$\sigma > 1 - \frac{1}{C_2(\log|t|)^{2/3}(\log\log|t|)^{1/3}}, \tag{1.1}$$

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where C_2 is a positive constant and are due to the method of Korobov [14] and Vinogradov [27], in which the main tool is an upper bound on $|\zeta(\sigma+it)|$ when σ is near to the line $\sigma=1$. Upper bounds of this form were first made explicit by Richert [21], who used the Korobov-Vinogradov method to prove that

$$|\zeta(\sigma+it)| \leq A|t|^{B(1-\sigma)^{3/2}} \log^{2/3}|t| \qquad |t| \geq 2,\ \frac{1}{2} \leq \sigma \leq 1 \tag{1.2}$$

with $B=100$ and A a certain absolute constant. Although smaller values for B were already known (see [15]), the first completely explicit bound of the form (1.2) is due to Cheng [4], with $A=175$ and $B=46$. This estimate was further improved in 2002 by Ford [8] to $A=76.2$ and $B=4.45$ for every $|t| \geq 3$ and $1/2 \leq \sigma \leq 1$.

Explicit bounds of the form (1.2) play a fundamental role in detecting explicit Korobov–Vinogradov zero-free regions for $\zeta(s)$. There are several results regarding the value of the constant C_2 in (1.1) and the current best known estimate is due to Mossinghoff, Trudgian and Yang [18], with $C_2 = 55.241$ for every $|t| \geq 3$ (see [5,6][1] for previous results).

In this paper, we will improve the values for both the constants A, B in (1.2) and, as a consequence, we find an improved Korobov-Vinogradov zero-free region for $\zeta(s)$. More precisely, denoting with $\zeta(s,u)$ the Hurwitz zeta function $\zeta(s,u)=\sum_{n=0}^{\infty}(n+u)^{-s}$ defined for every $\Re s>1$ and $0<u\leq 1$, we will prove the following result for both $\zeta(s)=\zeta(s,1)$ and a generic Hurwitz function $\zeta(s,u)$.

Theorem 1.1. *The following estimate holds for every $|t| \geq 3$ and $\frac{1}{2} \leq \sigma \leq 1$:*

$$\begin{aligned} |\zeta(\sigma+it)| &\leq A|t|^{B(1-\sigma)^{3/2}} \log^{2/3}|t| \\ \left|\zeta(\sigma+it,u)-u^{-s}\right| &\leq A|t|^{B(1-\sigma)^{3/2}} \log^{2/3}|t|, \qquad 0<u\leq 1, \end{aligned} \tag{1.3}$$

with $A=70.6995$ and $B=4.43795$.

The bound on $\zeta(s,u)$ found in Theorem 1.1 might be useful to bound Dirichlet L-functions due to the relation $L(s,\chi)=q^{-s}\sum_{m=1}^{q}\chi(m)\zeta(s,m/q)$, where χ is a Dirichlet character modulo q.

Although the new values $A=70.6995$ and $B=4.43795$ found in Theorem 1.1 are modest improvements on those found by Ford in [8] ($A=76.2$ and $B=4.45$ respectively), their importance relies on the fact that just a small improvement in both A and B can lead to improvements in several other results in analytic number theory. In particular, they have many applications in finding improved estimates for the Korobov–Vinogradov zero-free region and in estimating the error term in the prime number theorem, both in an effective and ineffective way.

Furthermore, improvements in A and B of the size found in Theorem 1.1 were already expected. In [8] (Section 8, point 1), Ford predicted that an optimization of the argument involving the Vinogradov integral would reduce B by less than 0.02, which is consistent with our new value for $B=4.43795$ found in Theorem 1.1.

As already mentioned, an immediate consequence of Theorem 1.1 is a new explicit zero free region for the Riemann zeta function.

Theorem 1.2. *There are no zeros of $\zeta(\sigma+it)$ for $|t| \geq 3$ and*

$$\sigma \geq 1 - \frac{1}{53.989(\log|t|)^{2/3}(\log\log|t|)^{1/3}}.$$

[1] An updated version of [6] is [7], where some errors have been corrected.

Theorem 1.1 has also an influence on asymptotically Korobov-Vinogradov zero-free regions of the form

$$\sigma \geq 1 - \frac{1}{c(\log|t|)^{2/3}(\log\log|t|)^{1/3}}, \tag{1.4}$$

where c is a positive constant for $|t|$ sufficiently large. More precisely, we get $c = 48.0718$, improving c on the current best value of 48.1588 due to Mossinghoff, Trudgian and Yang [18] (see [6] for previous results).

Theorem 1.3. *For sufficiently large $|t|$, there are no zeros of $\zeta(\sigma + it)$ with*

$$\sigma \geq 1 - \frac{1}{48.0718(\log|t|)^{2/3}(\log\log|t|)^{1/3}}. \tag{1.5}$$

As in [18], the proofs of Theorem 1.2 and Theorem 1.3 rely on some non-negative trigonometric polynomials. We recall that, for every $K \geq 2$, a K-th degree non-negative trigonometric polynomial $P_K(x)$ is defined as

$$P_K(x) = \sum_{k=0}^{K} b_k \cos(kx) \tag{1.6}$$

where b_k are constants such that $b_k \geq 0, b_1 > b_0$ and $P_K(x) \geq 0$ for all real x.

Finally, one can use Theorem 1.3 to improve on the error term in the prime number theorem. As per Ford [8], we get the following estimate for the error term in the prime number theorem

$$\pi(x) - \mathrm{li}(x) \ll x \exp\left\{-d(\log x)^{3/5}(\log\log x)^{-1/5}\right\},$$

with

$$d = \left(\frac{5^6}{2^2 \cdot 3^4 \cdot c^3}\right)^{1/5},$$

where c is the constant in (1.4). Using the new value for c found in Theorem 1.3 we obtain $d = 0.212579$, which is a slightly improvement on $d = 0.2123$ found by Mossinghoff, Trudgian and Yang [18].

As in [8], Theorem 1.1 follows from a uniform upper bound on the sum

$$S(N,t) := \max_{0<u\leq 1} \max_{N<R\leq 2N} \left| \sum_{N\leq n\leq R} \frac{1}{(n+u)^{it}} \right|, \tag{1.7}$$

where N is a positive integer and $t \geq N$. This upper bound is of the form

$$S(N,t) \leq cN^{1-1/(u\lambda^2)},$$

with $u, c > 0$ and $\lambda = \log t/\log N$. The strength of Ford's argument in [8] relies on some explicit estimates for both the Vinogradov integral and a quantity that counts the number of solutions of incomplete Diophantine systems that, combined with estimates for the exponential sum $S(N,t)$, give a completely explicit uniform upper bound on $S(N,t)$. Explicit bounds for the Vinogradov integral were further improved by Preobrazhenskiĭ [20] in 2011 and Steiner in 2019 [24] (see [9,23,25,1] for previous results).

We recall that the Vinogradov integral is defined as

$$J_{s,k}(P) = \int_{[0,1]^k} \left| \sum_{1\leq x\leq P} e\left(\alpha_1 x + \cdots + \alpha_k x^k\right) \right|^{2s} d\boldsymbol{\alpha} \tag{1.8}$$

where $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_k)$ and $e(z) = e^{2\pi i z}$, or equivalently, $J_{s,k}(P)$ is defined as the number of solutions of the simultaneous equations

$$\sum_{i=1}^{s} \left(x_i^j - y_i^j \right) = 0 \quad (1 \le j \le k); \quad 1 \le x_i, y_i \le P.$$

With regard to ineffective results involving the Vinogradov integral, lower bounds for $J_{s,k}(P)$ are well-known (see for example [26] for a proof) and are of the form

$$J_{s,k}(P) \gg_{s,k} \max \left\{ P^s, P^{2s - \frac{1}{2}k(k+1)} \right\}.$$

Furthermore, in two recent breakthroughs, the main conjecture in Vinogradov's Mean Value Theorem was proved by Wooley [30] in the case $k = 3$ and by Bourgain, Demeter and Guth [3] for $k > 3$ (see [31] for an alternative proof), that is, the estimate

$$J_{s,k}(P) \ll_{s,k,\epsilon} \max \left\{ P^{s+\epsilon}, P^{2s - \frac{1}{2}k(k+1)+\epsilon} \right\} \tag{1.9}$$

holds for every $s, k \in \mathbb{N}$ and $\epsilon > 0$. Also, applying the circle method one can show that the Vinogradov integral has an asymptotic behavior of the form

$$J_{s,k}(P) \sim C_{s,k} P^{2s - \frac{1}{2}k(k+1)}$$

for every $s > \frac{1}{2}k(k+1)$. However, although the exponent in the estimate for $J_{s,k}(P)$ is optimal, being ineffective it cannot be used to find explicit zero-free regions. An indication of the growth of the constant in the estimate of $J_{s,k}(P)$ was given recently by Steiner [24], who found $C_{s,k}$ of the order $k^{k^{O(\log(k)/\log(\lambda))}}$ for every $k \ge 3$, $s \ge \frac{5}{2}k^2 + k$, and $P \ge s^{10}$.

Regarding incomplete systems, we denote with $J_{s,k,h}(\mathscr{B})$ the number of solutions of the system

$$\sum_{i=1}^{s} \left(x_i^j - y_i^j \right) = 0 \quad (h \le j \le k); \quad x_i, y_i \in \mathscr{B},$$

where $\mathscr{B}$ is a suitable set. For our purpose, we will use the definition given by Ford in [8] $\mathscr{B} = \mathscr{C}(P, R)$ where $\mathscr{C}(P, R)$ is the set of integers $\le P$ composed only of prime factors in $(\sqrt{R}, R]$. Hence, $J_{s,k,h}(\mathscr{C}(P, R))$ is defined as

$$J_{s,k,h}(\mathscr{C}(P, R)) = \int_{[0,1]^t} |f(\boldsymbol{\alpha})|^{2s} d\boldsymbol{\alpha}, \tag{1.10}$$

where

$$f(\boldsymbol{\alpha}) = f(\boldsymbol{\alpha}; P, R) = \sum_{x \in \mathscr{C}(P,R)} e\left(\alpha_h x^h + \cdots + \alpha_k x^k \right)$$

and $\boldsymbol{\alpha} = (\alpha_h, \cdots, \alpha_k)$.

In his paper [8], Ford estimated $S(N, t)$ working separately and with different techniques on three different ranges of λ. More precisely, he considered the ranges $\lambda \le 87$, $87 \le \lambda \le 220$ and $\lambda \ge 220$, where the critical case happens to be around $\lambda = 87$. However, it is possible to shift the critical case to $\lambda = 84$, with a subsequent improved upper bound on $S(N, t)$ and improved values of A and B. This is made possible by introducing to Ford's method [8] some explicit bounds for the Vinogradov integral when $s \ll k^2$ due to

Tyrina [25]. This represents the main innovation of the paper. More precisely, in his paper [8] (p. 590), Ford stated a variant of his argument to find better estimates for the Vinogradov integral when k is small. These results, combined with Preobrazhenskiĭ's argument for $k \geq 90\ 000$ in [20] and Tyrina's explicit bounds [25], will give the following explicit bounds for $J_{s,k}$.

Theorem 1.4. *Let k and s be integers with $k \geq 90\ 000$ and*

$$0.138128k^2 \leq s \leq \frac{1}{2}k^2\left(0.6494 + \log\left(\frac{8k}{25D}\right) - \left(\frac{1-D/k}{2-D/k}\right)\frac{D}{k} + \frac{2.051}{k}\right),$$

with $1 \leq D \leq 0.4k$. Then

$$J_{s,k}(P) \leq k^{2.055k^3 - 0.414k^2 + 3nk} 1.06^{nk^2 + 2(n^2-n)k + 0.099912k^3} P^{2s - k(k+1)/2 + \Delta_s} \quad (P \geq 1),$$

where, denoting with $(s \bmod k)$ the unique integer u such that $0 \leq u < k$,

$$\begin{aligned}\Delta_s \leq \max\Bigg(&\frac{8}{25}k^2 \exp\left(0.6494 - \frac{2(s-k)}{k^2} - \left(\frac{1-D/k}{2-D/k}\right)\frac{D}{k} + \frac{2.051}{k} + \frac{2(s \bmod k)}{k^3}\right),\\ &Dk \exp\left(\frac{2(s \bmod k)}{k^2}\left(-1 + \frac{1}{k}\right)\right)\Bigg).\end{aligned} \tag{1.11}$$

Furthermore, if $k \geq 129$, there is an integer $s \leq \rho k^2$ such that for $P \geq 1$,

$$J_{s,k}(P) \leq k^{\theta k^3} P^{2s - \frac{1}{2}k(k+1) + 0.001k^2},$$

with

$$\begin{cases}
\rho = 3.177\ 207, & \theta = 2.409\ 292\ 660\ 6, & \text{if } 129 \leq k \leq 137\\
\rho = 3.177\ 527, & \theta = 2.395\ 29, & \text{if } 138 \leq k \leq 139\\
\rho = 3.178\ 551, & \theta = 2.391\ 67, & \text{if } 140 \leq k \leq 146\\
\rho = 3.178\ 871, & \theta = 2.382\ 59, & \text{if } 147 \leq k \leq 148\\
\rho = 3.181\ 869, & \theta = 2.379\ 29, & \text{if } 149 \leq k \leq 170\\
\rho = 3.184\ 127, & \theta = 2.353\ 34, & \text{if } 171 \leq k \leq 190\\
\rho = 3.192\ 950, & \theta = 2.333\ 13, & \text{if } 191 \leq k \leq 339\\
\rho = 3.196\ 497, & \theta = 2.243\ 52, & \text{if } 340 \leq k \leq 499\\
\rho = 3.205\ 501, & \theta = 1.777\ 75, & \text{if } 500 \leq k < 90\ 000\\
\rho = 3.208\ 630, & \theta = 2.177\ 20, & \text{if } k \geq 90\ 000.
\end{cases}$$

The values for ρ and θ are correct to ± 1 in the last place given.

Remark. The values for θ for $129 \leq k < 90\,000$ could all be given to the same precision as that for $129 \leq k \leq 137$ but it has no effect on the final result.

Using these new bounds for the Vinogradov integral we will prove the following result.

Theorem 1.5. *Suppose $N \geq 2$ is a positive integer, $N \leq t$ and set $\lambda = \frac{\log t}{\log N}$. Then*

$$S(N,t) \leq 8.7979 N^{1-1/(132.94357\lambda^2)}.$$

In proving both Theorem 1.4 and Theorem 1.5, we tried to make near-optimal choices for all parameters that are used in the argument. Hence, it seems unlikely that a substantial improvement in the constant B in Theorem 1.1 can be obtained via better choices of parameters alone.

An immediate corollary of Theorem 1.5 involving Dirichlet characters is the following one.

Corollary 1.6. *Suppose χ is a Dirichlet character modulo q, where $q \leq N$ and $2 \leq N \leq qt$. Then*

$$\max_{N<R\leq 2N} \left| \sum_{N<n\leq R} \chi(n) n^{-it} \right| \leq 9.7979 \frac{\phi(q)}{q} N e^{-\frac{\log^3(N/q)}{132.94357 \log^2 t}}.$$

The proof is the same as that of Corollary 2A in [8].

2. Background

We recall some preliminary results we will use later in the proof of the theorems.

First of all, given a fixed k, suppose $0 \leq d \leq k-1$ and T is a positive integer. The k-tuple of polynomials $\mathbf{\Psi} = (\Psi_1, \ldots, \Psi_k) \in \mathbb{Z}[x]^k$ is said to be of type (d,T) if Ψ_j is identically zero for $j \leq d$, and for some integer $m \geq 0$, when $j > d$, Ψ_j has degree $j-d$ with leading coefficient $\frac{j!}{(j-d)!} 2^m T$.

Then, let $K_{s,d}(P,Q;\mathbf{\Psi};q)$ be the number of solutions of

$$\sum_{i=1}^{k-d} \left(\Psi_j(z_i) - \Psi_j(w_i)\right) + q^j \sum_{i=1}^{s} \left(x_i^j - y_i^j\right) = 0 \quad (1 \leq j \leq k),$$

with $1 \leq z_i, w_i \leq P$ and $1 \leq x_i, y_i \leq Q$. Also, let $L_{s,d}(P,Q;\mathbf{\Psi};p,q,r)$ be the number of solutions of

$$\sum_{i=1}^{k-d} \left(\Psi_j(z_i) - \Psi_j(w_i)\right) + (pq)^j \sum_{i=1}^{s} \left(u_i^j - v_i^j\right) = 0 \quad (1 \leq j \leq k)$$

with $1 \leq z_i, w_i \leq P$, $z_i \equiv w_i \pmod{p^r}$ and $1 \leq u_i, v_i \leq Q$.

Lemma 2.1 *([8] Lemma 3.2'). Suppose k, r, d and s are integers with*

$$k \geq 4; \quad 2 \leq r \leq k; \quad 0 \leq d \leq r-1; \quad s \geq d+1.$$

Let M, P and Q be real numbers with

$$P^{1/(k+1)} \leq M \leq P^{1/r}; \quad 32 s^2 M < Q \leq P, \quad M \geq k.$$

Suppose q is a positive integer and $\mathbf{\Psi}$ is a system of polynomials of type (d,T) with $T \le P^d$. Denote by $\mathscr{P}$ the set of the k^3 smallest primes greater than M, and suppose $\mathscr{P} \subset (M, 2M]$. Then there is a system of polynomials $\mathbf{\Phi}$ of type (d,T) and a prime $p \in \mathscr{P}$ such that

$$K_{s,d}(P,Q;\mathbf{\Psi};q) \le 4k^3 k! p^{2s+(r-d)(r-d+1)/2} L_{s,d}(P,Q;\mathbf{\Phi};p,q,r).$$

Lemma 2.2 *([8] Lemma 3.3'). Suppose that $s \ge d, k \ge r \ge 2, d \le k-2, q \ge 1$, p is a prime and $\mathbf{\Phi}$ is a system of polynomials of type (d,T). Then there is a system of polynomials $\mathbf{\Upsilon}$ of type $(d+1,T')$ with $T \le T' \le PT$ such that*

$$L_{s,d}(P;Q;\mathbf{\Phi};p,q,r) \le (2P)^{k-d} \max\Big[k^{k-d} J_{s,k}(Q), 2p^{-r(k-d)} J_{s,k}(Q)^{(k-d-2)/(2(k-d-1))} \times K_{s,d+1}(P,Q;\mathbf{\Upsilon};pq)^{(k-d)/(2(k-d-1))}\Big].$$

As per Ford [8], Lemma 2.1 and Lemma 2.2 imply the following result.

Lemma 2.3. *Suppose $k \ge 26, 4 \le r \le k, k \le s \le k^3$ and*

$$J_{s,k}(Q) \le CQ^{2s-k(k+1)/2+\Delta} \quad (Q \ge 1).$$

Let j be an integer satisfying

$$2 \le j \le \frac{9}{10}r, \quad (j-1)(j-2) \le 2\Delta - (k-r)(k-r+1). \tag{2.1}$$

Define

$$\phi_J = \frac{1}{2r} + \frac{k^2+k+r^2-r-2\Delta-2rJ}{4r(k-J)}\phi_{J+1} \quad (1 \le J \le j-1),$$

and suppose r and j are chosen so that $\phi_i \ge 1/(k+1)$ for every i. Suppose

$$\frac{1}{3\log k} \le \omega \le \frac{1}{2}, \quad \eta = 1+\omega, \quad V = \max\left(e^{1.5+1.5/\omega}, \frac{18}{\omega}k^3 \log k\right).$$

If $P \ge V^{k+1}$, then

$$J_{s+k,k}(P) \le k^{3k} \eta^{4s+k^2} CP^{2(s+k)-k(k+1)/2+\Delta'}$$

where

$$\Delta' = \Delta(1-\phi_1) - k + \frac{1}{2}\phi_1\left(k^2+k+r^2-r\right).$$

We will use the definition of ϕ_J given in Lemma 2.3 for the range $k < 90\ 000$. However, when k becomes large, the influence that the new functions Φ_j have on estimating the Vinogradov's integral becomes negligible, as already pointed out by Ford in [8] (p. 590). Hence, for $k \ge 90\ 000$, we use the following result.

Lemma 2.4 *([8] Lemma 3.4). Suppose $k \ge 26, 4 \le r \le k, k \le s \le k^3$ and*

$$J_{s,k}(Q) \le CQ^{2s-k(k+1)/2+\Delta} \quad (Q \ge 1).$$

Let j be an integer satisfying the same relations as in (2.1). Then, define

$$\phi_j = \frac{1}{r}, \quad \phi_J = \frac{1}{2r} + \frac{k^2+k+r^2-r+J^2-J-2\Delta}{4kr}\phi_{J+1} \quad (1 \le J \le j-1),$$

and suppose r and j are chosen so that $\phi_i \ge 1/(k+1)$ for every i. Suppose

$$\frac{1}{3\log k} \le \omega \le \frac{1}{2}, \quad \eta = 1+\omega, \quad V = \max\left(e^{1.5+1.5/\omega}, \frac{18}{\omega}k^3\log k\right).$$

If $P \ge V^{k+1}$, then

$$J_{s+k,k}(P) \le k^{3k}\eta^{4s+k^2}CP^{2(s+k)-k(k+1)/2+\Delta'}$$

where

$$\Delta' = \Delta(1-\phi_1) - k + \frac{1}{2}\phi_1\left(k^2+k+r^2-r\right).$$

For a given k, r and Δ, we let $\delta_0(k,r,\Delta)$ be the value of Δ' coming from Lemma 2.3 or Lemma 2.4, where we take j maximal satisfying (2.1).

Lemma 2.5 *([8] Lemma 3.5). Let $k \ge 26$ and let ω, η and V be as in Lemma 2.3 or Lemma 2.4. Let $\Delta_1 = \frac{1}{2}k^2(1-1/k)$ and for $n \ge 1$, let r_n be an integer in $[4,k]$ satisfying*

$$\phi^*(k, r_n, \Delta_n) := \frac{2k}{2r_nk + 2\Delta_n - (k-r_n)(k-r_n+1)} \ge \frac{1}{k+1}.$$

Then set $\Delta_{n+1} = \delta_0(k, r_n, \Delta_n)$. If $n \le k^2$, then

$$J_{nk,k}(P) \le C_nP^{2nk-k(k+1)/2+\Delta_n} \quad (P \ge 1),$$

where $C_1 = k!$ and, for $n \ge 2$,

$$C_n = C_{n-1}\max\left[k^{3k}\eta^{4k(n-1)+k^2}, V^{(k+1)(\Delta_{n-1}-\Delta_n)}\right].$$

As already mentioned before, some explicit bounds for the Vinogradov's integral were found by Tyrina [25], and they give evident improvements when $s \ll k^2$. In her argument, Tyrina defines recursively sequences r_n, s_n, and Δ_{s_n}, with $s_1 = k$ and $\Delta_{s_1} = k$. More precisely, for $s \ge 1$ we take r_n to be the integer nearest to the number

$$\frac{2\Delta_{s_n} + k(k+1)}{2k+1}$$

and we define recursively

$$\Delta_{s_{n+1}} = \Delta_{s_n} + r_n - \frac{(2r_n-k-1)(2r_n-k)}{2r_n} - \frac{\Delta_{s_n}}{r_n}$$

where

$$s_{n+1} = k + r_1 + \cdots + r_n.$$

Then, we consider the function $x(y),\ k \le y < k(k+1)/2$:

$$x(y) = -y + 2k - (k+1)^2 \cdot \ln\left(1 - 2 \cdot (y-k)/\left(k^2 - k\right)\right).$$

This function increases monotonically and assumes the values from k to infinity as y increases from k to $k(k+1)/2$. The function $y = y(x)$ which is inverse to $x(y)$ is also monotonically increasing. Finally, let l_0 denote the integer $\lceil x\left(k^2/2\right)\rceil$, where

$$x\left(k^2/2\right) = -k^2/2 + 2k + (k+1)^2 \ln(k-1).$$

Tyrina proved the following result.

Theorem 2.6 *([25] Thm. 1). The mean value $J_{s,k}(P)$ satisfies the estimate*

$$J_{s,k}(P) \le D_s \cdot P^{2s-\kappa_s}$$

where $D_s = 2^{2s(k+sk^{-1})} k^{k+4s^2} s^{2(s-k)}$. Here

(1) *if $s = s_n, n = 1, 2, \ldots, s_n < l_0$, then $\kappa_s \ge \Delta_s \ge y(s)$;*
(2) *if $s = l_t = l_0 + kt, t = 0, 1, \ldots$, then*

$$\kappa_s \ge \Delta_s \ge \frac{k(k+1)}{2} - \frac{k+2}{2}\left(1 - \frac{1}{k}\right);$$

(3) *if $s_n < s < s_{n+1}$, then $\kappa_s \ge \Delta_s \ge \Delta_{s_{n+1}} \cdot s/s_{n+1}$;*
(4) *if $l_{t-1} < s < l_t, t = 1, 2, \ldots$, then $\kappa_s \ge \Delta_s \ge \Delta_{l_t} \cdot s/l_t$.*

New explicit bounds for $J_{s,k}$ were found by Ford in 2002 [8].

Theorem 2.7 *([8] Thm. 3). Let k and s be integers with $k \ge 1000$ and $2k^2 \le s \le \frac{k^2}{2}\left(\frac{1}{2} + \log\frac{3k}{8}\right)$. Then*

$$J_{s,k}(P) \le k^{2.055k^3 - 5.91k^2 + 3s} 1.06^{sk + 2s^2/k - 9.7278k^3} P^{2s - \frac{1}{2}k(k+1) + \Delta_s} \quad (P \ge 1)$$

where

$$\Delta_s = \frac{3}{8} k^2 e^{1/2 - 2s/k^2 + 1.7/k}.$$

Further, if $k \ge 129$, there is an integer $s \le \rho k^2$ such that for $P \ge 1$,

$$J_{s,k}(P) \le k^{\theta k^3} P^{2s - \frac{1}{2}k(k+1) + 0.001k^2}$$

with

$$(\rho, \theta) = \begin{cases} (3.214\ 32,\ 2.329\ 1) & (k \ge 200) \\ (3.217\ 34,\ 2.384\ 9) & (150 \le k \le 199) \\ (3.223\ 13,\ 2.418\ 3) & (129 \le k \le 149) \end{cases}. \tag{2.2}$$

In 2011, Preobrazhenskiĭ [20] improved the value of ρ in Theorem 2.7 when $k \geq 16\,000$ to $\rho = 3.213\,303$. Some further explicit bounds for the Vinogradov's integral were found also by Steiner [24] using Wooley's efficient congruencing method [28,29,31]. He proved that for every $k \geq 3$, $s \geq \frac{5}{2}k^2 + k$, and $P \geq s^{10}$, we have $J_{s,k}(P) \leq CP^{2s-\frac{1}{2}k(k+1)}$, where C is roughly $k^{k^{O(\log(k)/\log(\lambda))}}$. However, although the exponent of P is the optimal one, the constant C is far too large compared to that one found by Ford in Theorem 2.7, which is of order $k^{O(k^3)}$. In order to have a reasonable estimate for $J_{s,k}$, one should find a constant that is at most of order $k^{O(k^4)}$, since otherwise, after having taken the k^4-th root (this passage is required by the argument in Section 4, where the constant appearing in the upper bound for the Vinogradov integral will be elevated to the power of $1/(2rs)$, with both r and s of order $O(k^2)$), the constant cannot be controlled. Hence, as Steiner pointed out in his paper ([24], p. 359), the estimate he found for $J_{s,k}$ cannot be applied to Ford's argument.

In [8], Ford found also an upper bound for the incomplete systems defined by (1.10).

Theorem 2.8 *([8] Thm. 4). Let $k \geq 60$, $h \in [0.9k, k-2]$, $s \in [2t, \lfloor h/2 \rfloor t]$ and $P \geq \exp(Dk^2)$ for some $D \geq 10$. Assume that*

$$\eta \in \left(2k^{-3}, (2k)^{-1}\right], \qquad \frac{4\log k}{Dk^2\eta} \in \left[18k^{-1}, 0.4\right].$$

Then

$$J_{s,k,h}(\mathcal{C}(P, P^\eta)) \leq AP^E$$

where

$$A := \exp\left(\frac{s^2}{t} + \frac{10.5t\log^2 k}{Dk\eta^2} + s\left(\left(\eta^{-1}+h\right)\left(1-h^{-1}\right)^{s/t} - h\right)\log(10\eta)\right),$$
$$E := 2s - \frac{t(h+k)}{2} + \frac{t(t-1)}{2} + \frac{\eta s^2}{2t} + ht\exp\left(-\frac{s}{ht}\right).$$

Finally, we recall a few results found in [8] involving both $S(N,t)$ defined in (1.7) and some bounds for both $\zeta(s)$ and $\zeta(s,u)$ that we will use to prove Theorem 1.5 and Theorem 1.1 respectively.

Lemma 2.9 *([8] Lemma 5.1). Suppose k, r and s are integers ≥ 2, and h and g are integers satisfying $1 \leq h \leq g \leq k$. Let N be a positive integer, and M_1, M_2 be real numbers with $1 \leq M_i \leq N$. Let $\mathscr{B}$ be a nonempty subset of the positive integers $\leq M_2$. Then*

$$S(N,t) \leq 2M_1M_2 + \frac{t(M_1M_2)^{k+1}}{(k+1)N^k}$$
$$+ N\left(\frac{M_2}{|\mathscr{B}|}\right)^{1/r}\left((5r)^k M_2^{-2s}\lfloor M_1\rfloor^{-2r+k(k+1)/2} J_{r,k}(\lfloor M_1\rfloor) J_{s,g,h}(\mathscr{B}) W_h \dots W_g\right)^{1/2rs},$$

where

$$W_j = \min\left(2sM_2^j, \frac{2sM_2^j}{r\lfloor M_1\rfloor^j} + \frac{stM_2^j}{\pi jN^j} + \frac{4\pi j(2N)^j}{rt\lfloor M_1\rfloor^j} + 2\right) \quad (j \geq 1).$$

Lemma 2.10 *([8], Lemma 7.1). Suppose $\frac{1}{2} \leq \sigma \leq 1, 0 < u \leq 1, t \geq 3$ and $s = \sigma + it$. If either $\sigma \leq \frac{15}{16}$ or $t \leq 10^{100}$, then*

$$|\zeta(s)|, \left|\zeta(s,u) - u^{-s}\right| \leq 58.1t^{4(1-\sigma)^{3/2}}\log^{2/3} t.$$

Lemma 2.11 *([8], Lemma 7.2). If $s = \sigma + it$, $\frac{15}{16} \le \sigma \le 1, t \ge 10^{100}$ and $0 < u \le 1$, then*

$$\left| \zeta(s,u) - \sum_{0 \le n \le t} (n+u)^{-s} \right| \le 10^{-80}.$$

Lemma 2.12 *([8], Lemma 7.3). Suppose that $S(N,t) \le CN^{1-1/(D\lambda^2)} (1 \le N \le t)$ for positive constants C and D, where $\lambda = \frac{\log t}{\log N}$. Let $B = \frac{2}{9}\sqrt{3D}$. Then, for $\frac{15}{16} \le \sigma \le 1, t \ge 10^{100}$ and $0 < u \le 1$, we have*

$$|\zeta(s)| \le \left(\frac{C + 1 + 10^{-80}}{\log^{2/3} t} + 1.569 C D^{1/3} \right) t^{B(1-\sigma)^{3/2}} \log^{2/3} t$$

$$\left| \zeta(s,u) - u^{-s} \right| \le \left(\frac{C + 1 + 10^{-80}}{\log^{2/3} t} + 1.569 C D^{1/3} \right) t^{B(1-\sigma)^{3/2}} \log^{2/3} t.$$

3. Proof of Theorem 1.4

Let $s \ll k^2$. Due to the size of s, we are in the first or third cases of Theorem 2.6. However, since the third case gives an estimate for those s between two consecutive s_n for which we have an estimate in the first case, we can restrict ourselves to the first case. By Theorem 2.6, we know that

$$J_{s,k} \ll P^{2s - y(s)},$$

where $y(x)$ is the inverse function of $x(y)$, with

$$x(y) = -y + 2k - (k+1)^2 \log\left(1 - \frac{2(y-k)}{(k^2-k)}\right).$$

A direct computation gives the following explicit expression for $y(x)$:

$$y(x) = \frac{k(k+1)}{2} - (k+1)^2 W\left(\frac{k(k-1)}{2(k+1)^2} \exp\left(\frac{k}{2k+2} + \frac{x}{(k+1)^2} - \frac{2k}{(k+1)^2} \right) \right),$$

where W is the principal branch of the Lambert product-log function.

Furthermore, the function $x(y)$ is monotonically increasing in the interval $\left[n, \frac{n(n+1)}{2}\right)$. Therefore, to lower-bound $y(x)$, it suffices to upper-bound $x(y)$. We want to find s such that $y(s) \ge 0.101k^2$. This means that $s \le x(0.101k^2) \le 0.1247k^2$ for every $k \ge 50$.

However,

$$2s - 0.101k^2 \le 2s - \frac{k(k+1)}{2} + 0.4k^2, \qquad \forall k \ge 500, \quad s \ge 0.1247k^2. \tag{3.1}$$

It follows that, given

$$J_{s,k}(P) \ll P^{2s - \frac{k(k+1)}{2} + \Delta_s},$$

we have

$$\Delta_s \le 0.4k^2 \qquad \forall s \ge 0.1247k^2, \quad k \ge 500, \tag{3.2}$$

since Δ_s is decreasing in s, for every fixed k.

In order to prove the first part of Theorem 1.4, we follow Ford's method in [8] and we find an estimate for $J_{nk,k}$. By (3.2), we know that, given $k \geq 90\,000$ fixed, we have $\Delta_s \leq 0.4k^2$ for every $s \geq 0.1247k^2$. Hence, in particular, if we denote $n_0 = \lceil 0.1247k \rceil$, we have $\Delta_{n_0} := \Delta_{n_0 k} \leq 0.4k^2$. As a result, for the case $\Delta_{n-1} > k$, we follow Ford's method in Lemma 3.6 of [8] but instead of starting from Δ_1, we start from Δ_{n_0}.

Lemma 3.1. *For every* $1 \leq D \leq 0.4k$, $k \geq 90\,000$ *and*

$$0.138128k \leq n \leq \frac{1}{2}k\left(0.6494 + \log\left(\frac{8k}{25D}\right) - \left(\frac{1-D/k}{2-D/k}\right)\frac{D}{k} + \frac{2.051}{k}\right) + 1$$

we have

$$J_{nk,k}(P) \leq C_n P^{2nk-k(k+1)/2+\Delta_n}$$

with

$$\Delta_n \leq \max\left(\frac{8}{25}k^2 e^{0.6494-2n/k-\left(\frac{1-D/k}{2-D/k}\right)\frac{D}{k}+2.051/k}, Dk\right)$$

and

$$C_n \leq k^{2.055k^3-0.414k^2+3nk} 1.06^{nk^2+2(n^2-n)k+0.099912k^3}.$$

Proof. We take

$$r_n = \left\lfloor \sqrt{k^2+k-2\Delta_n} \right\rfloor$$

in Lemma 2.5, where ϕ_J are defined as in Lemma 2.4. Then, we define $\delta_n = \Delta_n/k^2$ for every n. Now, we fix $n \geq 2$ and we define $\delta = \delta_{n-1}$, $\delta' = \delta_n$, $\Delta = \Delta_{n-1}$, $\Delta' = \Delta_n$ and $r = r_{n-1}$.

If $\Delta_{n-1} \leq k$, then the upper bound for Δ_n follows trivially, since

$$\Delta_n \leq \Delta_{n-1} \leq k \leq Dk \qquad \forall 1 \leq D \leq 0.4k,\ k \geq 90\,000.$$

Hence, from now on, we just consider $\Delta_{n-1} > k$.

By (3.2), we know that, given $k \geq 90\,000$ fixed, we have $\Delta_s \leq 0.4k^2$ for every $s \geq 0.1247k^2$. Hence, in particular, if we denote $n_0 = \lceil 0.1247k \rceil$, we have $\Delta_{n_0} := \Delta_{n_0 k} \leq 0.4k^2$. At this point, we follow Ford's method in Lemma 3.6 of [8] but instead of starting from Δ_1, we start from Δ_{n_0}.

Let

$$y = 2\Delta - (k-r)(k-r+1), \qquad \phi^* = \phi^*(k,r,\Delta) = \frac{2k}{2rk+y}.$$

Using the definition of r_n, we have

$$\sqrt{k^2+k-2\Delta} - 1 \leq r = r_{n-1} = \lfloor \sqrt{k^2+k-2\Delta} \rfloor \leq \sqrt{k^2+k-2\Delta}.$$

Hence,

$$\begin{aligned}
\phi^* = \phi^*(k, r, \Delta) &= \frac{2k}{2rk + y} \\
&\geq \frac{2k}{2k\sqrt{k^2 + k - 2\delta k^2} + \delta k \left(2k - 1 + \frac{1}{2\delta k}\right)} \\
&= \frac{2k}{2k^2\sqrt{1 + \frac{1}{k} - 2\delta} + \delta k \left(2k - 1 + \frac{1}{2\delta k}\right)} \\
&\geq \frac{2k}{2k^2 \left(1 + \frac{1}{2k} - \delta\right) + \delta k \left(2k - 1 + \frac{1}{2\delta k}\right)} \\
&= \frac{2}{2k + 1 - 2\delta k + 2\delta k - \delta + \frac{1}{2k}} \\
&= \frac{2}{2k + 1 - \delta + \frac{1}{2k}} \\
&\geq \frac{1}{k + 1}
\end{aligned}$$

and so the hypotheses of Lemma 2.5 are satisfied.

As per Ford [8], we have

$$\theta_1 \leq \frac{2^{1-j}}{r} + \frac{2\phi^*}{kr} \leq \frac{0.071}{k^4 r} + \frac{2\phi^*}{kr}. \tag{3.3}$$

Since $\delta \leq 0.4$, denoting with

$$\alpha = \sqrt{k^2 + k - 2\delta k^2} - 1,$$

we have

$$\begin{aligned}
\phi^* &= \frac{2k}{2rk + y} \\
&\leq \frac{2k}{2k\alpha + 2\delta k^2 - (k - \alpha)(k - \alpha + 1)} \\
&\leq \frac{2}{(2 - \delta^2)\, k - \delta} - \frac{1}{k} + \frac{2k}{\sqrt{2k}(3 + 4k) + 2(-1 - 3k - k^2)} \\
&< \frac{1.08696}{k} + \frac{0.2363}{k^2} - \frac{1}{k} + \frac{2k}{\sqrt{2k}(3 + 4k) + 2(-1 - 3k - k^2)} \\
&\leq \frac{1.08696}{k} + \frac{0.2363}{k^2} - \frac{1}{k} \\
&= \frac{0.08696}{k} + \frac{0.2363}{k^2}
\end{aligned} \tag{3.4}$$

as

$$\frac{2k}{\sqrt{2k}(3 + 4k) + 2(-1 - 3k - k^2)} < 0 \qquad \forall k \geq 90\ 000.$$

It follows that

$$\theta_1 \leq \frac{0.071}{k^4 r} + \frac{0.08696}{k^2 r} + \frac{0.2363}{k^3 r} \leq \frac{0.08697}{k^2 r}.$$

As per Ford [8],

$$\begin{aligned}
\Delta' &= \Delta - k + \frac{\phi^* + \theta_1}{2}(2kr - y) \\
&\leq \Delta - k + \frac{1}{2}\phi^*(2kr - y) + \frac{0.043485}{k^2}\left(2k - \frac{y}{r}\right) \\
&= \Delta - k + \frac{1}{2}\phi^*(2kr - y) + \frac{0.043485}{k^2}\left(2k - \frac{1}{r}\left(2\Delta - (k-r)(k-r+1)\right)\right) \\
&\leq \Delta - k + \frac{1}{2}\phi^*(2kr - y) + \frac{0.043485}{k^2}\left(r - 1 + \frac{k^2 + k - 2\Delta}{r}\right) \\
&= \Delta - \frac{2ky}{2rk + y} + \frac{0.043485}{k^2}\left(r - 1 + \frac{k^2 + k - 2\Delta}{r}\right) \\
&= \Delta - 2k + 4k^2\frac{r}{2rk + y} + \frac{0.043485}{k^2}\left(r - 1 + \frac{k^2 + k - 2\Delta}{r}\right)
\end{aligned} \tag{3.5}$$

because

$$-k + \frac{1}{2}\phi^*(2kr - y) = -k + \frac{1}{2}\frac{2k}{2rk + y}(2kr - y) = -k + k\left(1 - \frac{2y}{2kr + y}\right) = -\frac{2ky}{2rk + y}.$$

As a function of the real variable r,

$$\frac{r}{2rk + y}$$

has a positive second derivative and a minimum at

$$r_0 = \sqrt{k^2 + k - 2\Delta}.$$

It follows that in the interval

$$\sqrt{k^2 + k - 2\Delta_{n-1}} - 1 \leq r \leq \sqrt{k^2 + k - 2\Delta}$$

the maximum occurs in

$$r_1 = \sqrt{k^2 + k - 2\Delta_{n-1}} - 1.$$

Hence,

$$\begin{aligned}
\frac{r}{2rk + y} &\geq \frac{r_0}{2r_0k + y} \\
&= \frac{\sqrt{k^2 + k - 2\delta k^2}}{2k(\sqrt{k^2 + k - 2\delta k^2}) + 2\delta k^2 - (k - \sqrt{k^2 + k - 2\delta k^2})(k - \sqrt{k^2 + k - 2\delta k^2} + 1)} \\
&\geq \frac{1 - \delta}{2k}
\end{aligned} \tag{3.6}$$

and

$$\begin{aligned}
\frac{r}{2rk + y} &\leq \frac{r_1}{2r_1k + y} \\
&= \frac{\sqrt{k^2 + k - 2\delta k^2} - 1}{2k(\sqrt{k^2 + k - 2\delta k^2} - 1) + 2\delta k^2 - (k - \sqrt{k^2 + k - 2\delta k^2} + 1)(k - \sqrt{k^2 + k - 2\delta k^2} + 1 + 1)}.
\end{aligned}$$

We want to estimate the quantity

$$\frac{r_1}{2r_1k+y}$$

using Taylor expansion for $k \to +\infty$. We observe that

$$\lim_{k\to\infty} k \cdot \frac{r_1}{2r_1k+y} = \frac{\sqrt{1-2\delta}}{-2+4\sqrt{1-2\delta}+4\delta}.$$

Furthermore, for all $\delta \in (0, 0.4)$ we have

$$\frac{\sqrt{1-2\delta}}{-2+4\sqrt{1-2\delta}+4\delta} < \frac{1-\delta}{2-\delta^2}.$$

Now,

$$\frac{\sqrt{1-2\delta}}{-2+4\sqrt{1-2\delta}+4\delta} = \frac{1}{2} - \frac{\delta}{2} + \frac{\delta^2}{4} - \frac{\delta^3}{4} + \frac{\delta^4}{16} - \frac{3\delta^5}{16} + O(\delta^6)$$

and

$$\frac{1-\delta}{2-\delta^2} = \frac{1}{2} - \frac{\delta}{2} + \frac{\delta^2}{4} - \frac{\delta^3}{4} + \frac{\delta^4}{8} - \frac{\delta^5}{8} + O(\delta^6).$$

It follows that

$$\frac{\sqrt{1-2\delta}}{-2+4\sqrt{1-2\delta}+4\delta}\frac{1}{k} \leq \left(\frac{1-\delta}{2-\delta^2} - \frac{\delta^4}{16} - \frac{\delta^5}{16}\right)\frac{1}{k}. \tag{3.7}$$

Now, we repeat the process for the term of second order. We have

$$\lim_{k\to\infty} k^2 \cdot \left(\frac{r_1}{2r_1k+y} - \frac{\sqrt{1-2\delta}}{-2+4\sqrt{1-2\delta}+4\delta}\frac{1}{k}\right) = \frac{-1+\sqrt{1-2\delta}+2\delta}{4(-1+2\sqrt{1-2\delta}+2\delta)^2}.$$

As before, using the Taylor expansion in $\delta = 0$ ($\delta > 1/k \to 0$ for $k \to +\infty$), we have

$$\frac{-1+\sqrt{1-2\delta}+2\delta}{4(-1+2\sqrt{1-2\delta}+2\delta)^2}\frac{1}{k^2} \leq \left(\frac{4\delta}{(2-\delta^2)^2} - \frac{3\delta}{4}\right)\frac{1}{k^2}. \tag{3.8}$$

However, in order to have an upper bound for

$$\frac{r_1}{2r_1k+y}$$

a direct computation shows that we need to use the slightly weaker bound

$$\frac{-1+\sqrt{1-2\delta}+2\delta}{4(-1+2\sqrt{1-2\delta}+2\delta)^2}\frac{1}{k^2} \leq \left(\frac{4\delta}{(2-\delta^2)^2} - \frac{14\delta}{25}\right)\frac{1}{k^2}, \tag{3.9}$$

where we have $-14\delta/25$ instead of $-3\delta/4$. Using the upper bounds found in (3.7) and (3.9) we obtain

$$\frac{r}{2rk+y} \leq \left(\frac{1-\delta}{2-\delta^2} - \frac{\delta^4}{16} - \frac{\delta^5}{16}\right)\frac{1}{k} + \left(\frac{4\delta}{(2-\delta^2)^2} - \frac{14\delta}{25}\right)\frac{1}{k^2}. \tag{3.10}$$

From (3.5) and (3.10), it follows that

$$\begin{aligned}
\delta' &\le \delta - \frac{2}{k} + 4\left(\frac{1-\delta}{2-\delta^2} - \frac{\delta^4}{16} - \frac{\delta^5}{16}\right)\frac{1}{k} + \left(\frac{4\delta}{(2-\delta^2)^2} - \frac{14\delta}{25}\right)\frac{1}{k^2} \\
&\quad + \frac{0.043485}{k^4}\left(r - 1 + \frac{k^2+k-2\Delta}{r}\right) \\
&= \delta\left(1 - \frac{4-2\delta}{(2-\delta^2)k} - \frac{\delta^3}{4k} - \frac{\delta^4}{4k} + \frac{4}{(2-\delta^2)^2k^2} - \frac{14}{25k^2}\right) \\
&\quad + \frac{0.043485}{k^4}\left(\sqrt{k^2+k-2\delta k^2} - 1 + \frac{k^2+k-2\delta k^2}{\sqrt{k^2+k-2\delta k^2}-1}\right) \\
&\le \delta\left(1 - \frac{4-2\delta}{(2-\delta^2)k} - \frac{\delta^3}{4k} - \frac{\delta^4}{4k} + \frac{4}{(2-\delta^2)^2k^2} - \frac{14}{25k^2}\right) + \frac{0.08697(1-\delta)}{k^3} \\
&\le \delta\left(1 - \frac{2-\delta}{2-\delta^2}\left(\frac{2}{k} + \frac{\delta^3}{4k} + \frac{\delta^4}{4k} - \frac{32}{21k^2} + \frac{14}{25k^2}\right)\right) + \frac{0.08697}{k^3}\frac{2-\delta}{2-\delta^2} \\
&= \delta\left(1 - \frac{2-\delta}{2-\delta^2}\left(\frac{2}{k} + \frac{\delta^3}{4k} + \frac{\delta^4}{4k} - \frac{506}{525k^2} - \frac{0.08697}{k^3\delta}\right)\right) \\
&\le \delta\left(1 - \frac{2-\delta}{2-\delta^2}\left(\frac{2}{k} + \frac{1}{4k^4} + \frac{1}{4k^5} - \frac{506}{525k^2} - \frac{0.08697}{k^3\delta}\right)\right),
\end{aligned}$$

where in the last passage we used the fact that $\delta > 1/k$.

Now we define

$$\beta = \frac{2}{k} - \frac{506}{525k^2} + \frac{1}{4k^4} + \frac{1}{4k^5}, \qquad \beta' = \beta - \frac{c}{\delta}, \qquad c = \frac{0.08697}{k^3}$$

and let

$$\delta'' = \delta\left(1 - \frac{2-\delta}{2-\delta^2}\beta'\right).$$

Since $y + \log y + \log(2-y)$ is increasing on $\left(0, \frac{1}{2}\right]$, and hence in $(0, 0.4]$, we have

$$\begin{aligned}
\delta' + \log\delta' + \log(2-\delta') &\le \delta'' + \log\delta'' + \log(2-\delta'') \\
&= \delta + \log\delta + \log(2-\delta) \\
&\quad - \frac{2\delta-\delta^2}{2-\delta^2}\beta' + \log\left[\left(1 - \frac{2-\delta}{2-\delta^2}\beta'\right)\left(\frac{2-\delta''}{2-\delta}\right)\right].
\end{aligned}$$

As per Ford [8], given

$$T = -\frac{2\delta-\delta^2}{2-\delta^2}\beta' + \log\left(1 - \frac{2-\delta}{2-\delta^2}\beta'\right) + \log\left(\frac{2-\delta''}{2-\delta}\right),$$

we have

$$\begin{aligned}
T &\le -\beta' - \frac{(\beta')^2}{2(2-\delta^2)^2}\left((2-\delta)^2 + \delta^2\right) + \frac{(\beta')^3}{3(2-\delta^2)^3}\left(-(2-\delta)^3 + \delta^3\right) \\
&\le -\beta' - \frac{2}{5}(\beta')^2 \\
&\le -\beta - \frac{2}{5}\beta^2 + \frac{c(1+0.8\beta)}{\delta}.
\end{aligned}$$

Hence, using an iterative argument we get

$$\begin{aligned}\delta_n + \log\delta_n + \log(2-\delta_n) \leq & \delta_{n_0} + \log\delta_{n_0} + \log(2-\delta_{n_0}) - (n-n_0)\left(\beta + 0.4\beta^2\right) \\ & + c(1+1.6/k)\left(\frac{1}{\delta_{n_0}} + \ldots + \frac{1}{\delta_{n-1}}\right).\end{aligned}$$

Since we are working with $\Delta_n > k$, that is $\delta > 1/k$, and we found the inequality

$$\delta' \leq \delta\left(1 - \frac{2-\delta}{2-\delta^2}\left(\frac{2}{k} + \frac{1}{4k^4} + \frac{1}{4k^5} - \frac{506}{525k^2} - \frac{0.08697}{k^3\delta}\right)\right), \tag{3.11}$$

we have

$$\delta_{i+1} \leq \delta_i(1-\alpha), \quad \alpha = 0.869565(\beta - kc).$$

It follows that

$$c(1+1.6/k)\left(\frac{1}{\delta_{n_0}} + \ldots + \frac{1}{\delta_{n-1}}\right) \leq \frac{c(1+1.6/k)}{\alpha\delta_{n-1}} \leq \frac{0.051}{k}.$$

Therefore

$$\delta_n \leq \frac{\delta_{n_0}(2-\delta_{n_0})\,e^{\delta_{n_0}}}{(2-\delta_n)\,e^{\delta_n}} e^{-(n-n_0)(\beta+0.4\beta^2)+0.051/k}$$

Now,

$$\beta + 0.4\beta^2 \geq \frac{2}{k} - \frac{506}{525k^2} + \frac{1}{4k^4} + \frac{1}{4k^5} + \frac{0.4}{k^2}\left(2 - \frac{506}{525\cdot 90\,000}\right)^2 \geq \frac{2}{k}.$$

Furthermore

$$\frac{e^{-\delta_n}}{2-\delta_n} \leq \frac{1}{2} e^{\delta_n/(2-\delta_n)-\delta_n}$$

and, since $\delta_n = \Delta_n/k^2$, and $\delta_n \leq 0.4$ for $k \geq 90\,000$, if we write $\delta_n = D/k$, where $1 \leq D \leq 0.4k$, we have

$$\frac{e^{-\delta_n}}{2-\delta_n} \leq \frac{1}{2} e^{\delta_n/(2-\delta_n)-\delta_n} = \frac{1}{2} e^{-\left(\frac{1-D/k}{2-D/k}\right)\frac{D}{k}}.$$

Also, using the fact that $\delta_{n_0} \leq 0.4$, we have

$$\delta_{n_0}(2-\delta_{n_0})\,e^{\delta_{n_0}} \leq 0.4\cdot(2-0.4)\,e^{0.4} = \frac{16}{25}e^{0.4}.$$

Furthermore, for $k \geq 90\,000$, we have

$$e^{2n_0/k} \leq e^{0.2494+2/k}.$$

It follows that

$$\delta_n \leq \frac{8}{25} e^{0.6494-2n/k-\left(\frac{1-D/k}{2-D/k}\right)\frac{D}{k}+2.051/k}$$

and so

$$\Delta_n \leq \frac{8}{25}k^2 e^{0.6494-2n/k-\left(\frac{1-D/k}{2-D/k}\right)\frac{D}{k}+2.051/k}.$$

Now we shift our attention on the constant. As per Ford [8], to bound the constants C_n, we choose $\omega = 0.06 > 1/(3\log k)$ in Lemma 2.5, so that

$$V^{k+1} = \left(300k^3\log k\right)^{k+1} \leq k^{4.11k} =: W$$

for every $k \geq 90\ 000$. Then,

$$W^{\Delta_{n-1}-\Delta_n} > k^{3k}1.06^{4k(n-1)+k^2} \quad (n \leq 0.138k+1), \tag{3.12}$$

since it was proved by Ford in [8] for the wider range $n \leq 1.97k+1$.

Now, let $n_1 = \lfloor 0.138k \rfloor + 1$. By (3.12) and Lemma 2.5,

$$C_{n_1} \leq W^{\Delta_1-\Delta_{n_1}}k! \leq W^{k^2/2-\Delta_{n_1}}$$

and, for $n > n_1$,

$$C_n \leq k^{3k}1.06^{4k(n-1)+k^2}W^{\Delta_{n-1}-\Delta_n}C_{n-1}.$$

Iterating this last inequality gives, for $n > n_1$,

$$\begin{aligned} C_n &\leq W^{k^2/2}k^{3k(n-n_1)}1.06^{(n-n_1)k^2+4k(n_1+\ldots+n-1)} \\ &\leq W^{k^2/2}k^{3k(n-0.138k)}1.06^{(n-0.138k)k^2+2k(n^2-n-(0.138k)^2+0.138k)} \\ &\leq k^{2.055k^3-0.414k^2+3nk}1.06^{nk^2+2(n^2-n)k+0.099912k^3}. \end{aligned}$$

This concludes the proof of the lemma. □

Remark. In Lemma 3.1, contrary to Lemma 3.6 in [8], we cannot take $2k$ as a lower bound for the range of n for which the lemma holds. Indeed, the upper bound for n, that is $\frac{1}{2}k\Big(0.6494+\log\left(\frac{8k}{25D}\right)-\left(\frac{1-D/k}{2-D/k}\right)\frac{D}{k}+\frac{2.051}{k}\Big)+1$, is a decreasing function in D. As a result, this upper bound reaches the minimum when, for $k \geq 90\ 000$ fixed, $D = 0.4k$. Substituting $D = 0.4k$ in the above expression, one gets $n \leq 2.0255+0.138128k$, which is less than $2k$ for $k \geq 90\ 000$. Hence, in order to have an estimate for Δ_n and C_n which is uniform for every $1 \leq D \leq 0.4k$ and $k \geq 90\ 000$, we took $0.138128k$ as lower bound for n, in order to ensure that for every D and k there exists always at least a value of n in the range given in the hypotheses of Lemma 3.1. Also, the choice $0.138128k$ as lower bound for n is admissible, as in Lemma 3.1 the starting point is $n_0 = \lceil 0.1247k \rceil$ which is less than $0.138128k$ for $k \geq 90\ 000$.

Now, we will use Lemma 3.1 to prove the first part of Theorem 1.4.

Given $k \geq 90\ 000$, every admissible s such that $s \not\equiv 0 \pmod{k}$, is of the form $s = nk+u$, where $0 < u < k,\ u \equiv s \bmod k$, and

$$0.138128k \leq n \leq \frac{1}{2}k\left(0.6494+\log\left(\frac{8k}{25D}\right)-\left(\frac{1-D/k}{2-D/k}\right)\frac{D}{k}+\frac{2.051}{k}\right).$$

Using Hölder's inequality, we get

$$J_{s,k}(P) = \int\limits_{[0,1]^k} |S|^{2nk+2u} d\boldsymbol{\alpha} \leq \left(\int\limits_{[0,1]^k} |S|^{2nk} d\boldsymbol{\alpha} \right)^{1-u/k} \left(\int\limits_{[0,1]^k} |S|^{2nk} |S|^{2k} d\boldsymbol{\alpha} \right)^{u/k},$$

where $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_k)$ and

$$S := \sum_{1 \leq x \leq P} e\left(\alpha_1 x + \cdots + \alpha_k x^k\right).$$

By Lemma 3.1, we have

$$J_{s,k}(P) \leq k^{2.055k^3 - 0.414k^2 + 3nk} 1.06^{nk^2 + 2(n^2-n)k + 0.099912k^3} P^{2s-k(k+1)/2+\Delta}$$

where

$$\Delta \leq \max\left(\frac{8}{25} k^2 e^{0.6494 - 2n/k - \left(\frac{1-D/k}{2-D/k}\right)\frac{D}{k} + 2.051/k}, Dk \right) \times \left(1 - \frac{u}{k} + \frac{u}{k} e^{-2/k}\right).$$

Further,

$$1 - \frac{u}{k} + \frac{u}{k} e^{-2/k} \leq 1 - \frac{2u}{k^2} + \frac{2u}{k^3} \leq e^{-2u/k^2 + 2u/k^3}.$$

Hence,

$$\begin{aligned}
\Delta \leq & \max\left(\frac{8}{25} k^2 \exp\left(0.6494 - \frac{2(nk+u-k)}{k^2} - \left(\frac{1-D/k}{2-D/k}\right)\frac{D}{k} + \frac{2.051}{k} + \frac{2u}{k^3}\right), \right. \\
& \left. Dk \exp\left(-\frac{2u}{k^2} + \frac{2u}{k^3}\right)\right) \\
= & \max\left(\frac{8}{25} k^2 \exp\left(0.6494 - \frac{2(s-k)}{k^2} - \left(\frac{1-D/k}{2-D/k}\right)\frac{D}{k} + \frac{2.051}{k} + \frac{2(s \bmod k)}{k^3}\right), \right. \\
& \left. Dk \exp\left(\frac{2(s \bmod k)}{k^2}\left(-1 + \frac{1}{k}\right)\right)\right).
\end{aligned}$$

This completes the first part of Theorem 1.4.

We now turn to the second part of the theorem. For $k < 90\,000$, we follow Ford's argument for proving Theorem 2.7 in [8]. Running Program 1 in [2] with ϕ_J defined as in Lemma 2.3, we get the desired bounds for k in the ranges

$$[129, 137],\ [138, 139],\ [140, 146],\ [147, 148],\ [149, 170],\ [171, 190],\ [191, 339],\ [340, 499].$$

For $500 \leq k < 90\,000$ we use Program 2 in [2], with ϕ_J defined as in Lemma 2.3. Program 2 differs from Program 1 in the definition of the variable del0, as, due to (3.2), we know that, for $k \geq 500$, $\Delta_n \leq 0.4k^2$ for every $n \geq n_0 = \lceil 0.1247k \rceil$. Hence, in order to initialize del0, we start from $\Delta_{n_0} \leq 0.4k^2$ instead of starting from $\Delta_1 \leq \frac{1}{2}k^2(1 - \frac{1}{k})$.

For $k \geq 90\,000$, we follow Preobrazhenskiĭ's argument [20]. Taking $D = 0.001k$ in Lemma 3.1, we observe that for every $k \geq 90\,000$,

$$2k \leq n \leq \frac{1}{2} k \left(0.6494 + \log\left(\frac{8k}{25D}\right) - \left(\frac{1-D/k}{2-D/k}\right)\frac{D}{k} + \frac{2.051}{k}\right).$$

Hence, since from now on we will work with $D = 0.001k$, we will restrict the range of n in Lemma 3.1 to

$$2k \leq n \leq \frac{1}{2}k\left(0.6494 + \log\left(\frac{8k}{25D}\right) - \left(\frac{1-D/k}{2-D/k}\right)\frac{D}{k} + \frac{2.051}{k}\right) + 1. \tag{3.13}$$

Choosing

$$n = \left\lceil \frac{1}{2}k\left(0.6494 + \log\left(\frac{8k}{25D}\right) - \left(\frac{1-D/k}{2-D/k}\right)\frac{D}{k} + \frac{2.051}{k}\right)\right\rceil, \tag{3.14}$$

we have, by Lemma 3.1, that

$$\begin{aligned}\Delta_n &\leq \max\left(\frac{8}{25}k^2 e^{0.6494-2n/k-\left(\frac{1-0.001}{2-0.001}\right)0.001+2.051/k}, 0.001k^2\right)\\ &\leq \max\left(\frac{8}{25}k^2 e^{-\log\left(\frac{8}{25\cdot 0.001}\right)}, 0.001k^2\right)\\ &= 0.001k^2,\end{aligned}$$

where we used the fact that

$$-\frac{2n}{k} \leq -\left(0.6494 + \log\left(\frac{8k}{25D}\right) - \left(\frac{1-D/k}{2-D/k}\right)\frac{D}{k} + \frac{2.051}{k}\right).$$

Hence, for every $k \geq 90\ 000$ fixed, there exists $s \leq \rho k^2$ such that

$$J_{s,k}(P) \leq k^{\theta k^3} P^{2s-\frac{1}{2}k(k+1)+0.001k^2},$$

where

$$\rho \leq \frac{nk}{k^2} \leq \frac{1}{2}\left(0.6494 + \log\left(\frac{8k}{25D}\right) - \left(\frac{1-D/k}{2-D/k}\right)\frac{D}{k} + \frac{2.051}{k}\right) + \frac{1}{k} \leq 3.20863$$

and $\theta = 2.17720$ (the value of θ follows from a uniform upper bound for the constant

$$k^{2.055k^3-0.414k^2+3nk}1.06^{nk^2+2(n^2-n)k+0.099912k^3}$$

found in the first part of Theorem 1.4 which holds for every $k \geq 90\ 000$, and n in the range (3.13)). This estimate holds since, using our choice (3.14) for n, we have $s = nk \leq \rho k^2$ and $\Delta_n \leq 0.001k^2$.

Remark. The starting point $\Delta_{n_0} \leq 0.4k^2$ with $n_0 = \lceil 0.1247k \rceil$ is nearly the optimal choice. Indeed, taking $\Delta_{n_0} \leq ck^2$ with $c < 0.4$ would imply a greater value for n_0. A direct computation shows that, although the constant $\frac{8}{25}$ in the estimate of Δ_n would decrease, the quantity $e^{0.6494}$ would increase, leading to an overall estimate for Δ_n which is worse than that one found by Ford in Lemma 3.6 of [8] when $\Delta_n > k$.

Furthermore, the lower bound for k in (3.1) is optimal, as the inequality in (3.1) is no longer valid for $k < 500$. Finally, the value $0.101k^2$ in (3.1) is also nearly optimal, as this inequality is satisfied only with $2s - ak^2$ and $a > 0.1$.

4. Proof of Theorem 1.5

We will consider the cases $\lambda \leq 84$ and $\lambda \geq 84$ separately.

4.1. Case $\lambda \geq 84$

We combine Ford's method [8] with the new estimates for the Vinogradov's integral found in Theorem 1.4 to obtain an improved upper bound for $S(N,t)$ in Lemma 2.9.

As in [8], we make the following assumptions:

$$\lfloor M_1 \rfloor \geq M_2 \geq 100g, \quad s \leq 2^g, \quad r \geq 13g, \quad r \geq s, \quad g \geq h \geq 3.$$

Also, we define

$$M_1 = N^{\mu_1}, \quad M_2 = N^{\mu_2}, \quad \mu_1 > \mu_2$$

and

$$\phi = g/\lambda, \quad \gamma = h/\lambda, \quad 1 \leq \gamma \leq \frac{1}{1-\mu_2} < \frac{1}{1-\mu_1} \leq \phi \leq \frac{1}{1-\mu_1-\mu_2}. \tag{4.1}$$

Following the proof of Theorem 2 in [8] for λ large, using the notation of Lemma 2.9, we have

$$W_h \cdots W_g \leq 2^{g^2} M_2^{h+(h+1)+\cdots+g} N^{-H} \tag{4.2}$$

where

$$\begin{aligned} H \geq & \lambda^2 \left(\phi + \gamma - \frac{\gamma^2}{2} - \frac{1-\mu_1-\mu_2}{2}\phi^2 - \frac{2-\mu_1-\mu_2}{2\,(1-\mu_1)\,(1-\mu_2)} \right) \\ & + \lambda \left(\frac{\gamma}{2} - \frac{\phi}{2}\,(1-\mu_1-\mu_2) \right) - \frac{2-\mu_1-\mu_2}{8} \\ =: & H_2\lambda^2 + H_1\lambda - H_0. \end{aligned} \tag{4.3}$$

At this point, we shall take the near-optimal values

$$\mu_1 = 0.1905, \quad \mu_2 = 0.1603, \quad k = \left\lfloor \frac{\lambda}{1-\mu_1-\mu_2} + 0.000003 \right\rfloor \geq 129, \quad r = \left\lfloor \rho k^2 + 1 \right\rfloor, \tag{4.4}$$

where ρ is taken from Theorem 1.4.

Furthermore, we choose $Y = 288$ and we want to estimate $S(N,t)$ when $N \geq e^{Y\lambda^2}$, since otherwise for $N \leq e^{Y\lambda^2}$ Theorem 1.5 follows trivially:

$$S(N,t) \leq N \leq e^{Y/132.94357} N^{1-1/132.94357\lambda^2} \leq 8.7979 N^{1-1/132.94357\lambda^2}.$$

Finally, we consider

$$105 \leq g \leq 1.254\lambda \tag{4.5}$$

and the following bounds for the quantity k/λ:

$$k_0 := \frac{1}{0.6492} - \frac{0.999997}{\lambda} \leq \frac{k}{\lambda} \leq \frac{1}{0.6492} + \frac{0.000003}{\lambda} =: k_1.$$

First of all, by (4.4) and Theorem 1.4 we have

$$\lfloor M_1 \rfloor^{-2r+\frac{1}{2}k(k+1)}\, J_{r,k}\left(\lfloor M_1 \rfloor\right) \leq C_1 M_1^{0.001k^2}, \tag{4.6}$$

where $C_1 = k^{\theta k^3}$ and θ comes from Theorem 1.4.

Since $N \geq e^{Y\lambda^2}$, from (4.4) and (4.5) we have $M_2 \geq e^{\mu_2 Y\lambda^2} \geq e^{0.1019Yg^2}$. Let $D = 0.1019Y = 29.3472$ and $\eta = \frac{1}{\xi g^{3/2}}$, where $3 \leq \xi \leq 6$. By the assumption on g, (4.5), the hypotheses of Theorem 2.8 are satisfied if we take $P = M_2$ and $k = g$. Hence, by Theorem 2.8,

$$J_{s,g,h}\left(\mathscr{C}\left(M_2, M_2^{\eta}\right)\right) \leq C_2 P^{2s-\frac{t}{2}(h+g)+E_2}, \tag{4.7}$$

where

$$\begin{aligned} E_2 &= \frac{1}{2}t(t-1) + \frac{\eta s^2}{2t} + ht\exp\left\{-\frac{s}{ht}\right\} \\ \log C_2 &= \frac{s^2}{t} + \frac{10.5\xi^2 t g^2 \log^2 g}{D} - s\left(\left(\xi g^{3/2} + h\right)(1-1/h)^{s/t} - h\right)\log\left(\xi g^{3/2}/10\right). \end{aligned}$$

Since by the hypotheses of Theorem 2.8 and the range (4.5) for g we have

$$R = M_2^{\eta} \geq e^{Dg^2\eta} \geq g^{10} > 6^{25},$$

following exactly the argument used by Ford in the proof of Theorem 2 in [8] for λ large with $\delta = \frac{1}{25}$ instead of $\delta = \frac{1}{26}$ we get

$$\begin{aligned} \frac{M_2}{|\mathscr{C}(M_2, R)|} &\leq (\log R)\left(1.0417\xi g^{3/2} + 1\right)\left(\frac{26.0417\xi g^{3/2}}{2.5}\right)^{1.0417\xi g^{3/2}} \\ &\leq (\log N)C_3 \leq C_3 N^{E_3}, \end{aligned}$$

where

$$\begin{aligned} C_3 &= \left(10.4167\xi g^{3/2}\right)^{1.0417\xi g^{3/2}} \\ E_3 &= \frac{\log\left(Y\lambda^2\right)}{Y\lambda^2}. \end{aligned}$$

By (4.4), we have

$$(5r)^k \leq \left(38.2\lambda^2\right)^{1.55\lambda} \leq \lambda^{4.65\lambda} \tag{4.8}$$

and

$$r \geq 7.6\lambda^2 \geq \lambda^2.$$

It follows that

$$\frac{E_3}{r} \leq \frac{\log\left(Y\lambda^2\right)}{7.6Y\lambda^4} \leq \frac{\log\left(Y\lambda^2\right)}{Y\lambda^4}. \tag{4.9}$$

Lemma 4.1. *For $84 \leq \lambda \leq 220$ and $N \geq e^{288\lambda^2}$ we have the following estimate*

$$S(N,t) \leq 8.7979 N^{1-1/132.94357\lambda^2}.$$

Proof. From Lemma 2.9, (4.2), (4.4), (4.6), (4.7), (4.8) and (4.9) we have

$$\begin{aligned} S(N,t) &\leq \left(C_3^{\frac{1}{r}}\left(\lambda^{4.65\lambda}C_1C_2\right)^{\frac{1}{2rs}}\right)N^{1+E} + 2N^{0.36} + \frac{1}{k}N^{1-0.0000019476} \\ E &= \frac{\log\left(Y\lambda^2\right)}{Y\lambda^4} + \frac{1}{2rs}\left(-H + 0.001\mu_1k^2 + \mu_2E_2\right). \end{aligned} \tag{4.10}$$

Now, we choose

$$\xi = 3.612\ 381, \qquad \sigma = 0.330\ 201.$$

Furthermore, taking $s = \lfloor \sigma ht \rfloor + 1$, we make the same assumptions as in Lemma 5.3 of [8]

$$g = \left\lfloor \frac{\lambda}{1-\mu_1} \right\rfloor + 1 + a, \quad h = \left\lfloor \frac{\lambda}{1-\mu_2} \right\rfloor - b, \quad t = g - h + 1, \quad a, b \in \{0,1\},$$

where g satisfies (4.5). Then, we bound the exponent of N in each interval $\lambda \in I = [\lambda_1, \lambda_2)$ where each of the quantities $m_1 = \left\lfloor \frac{\lambda}{1-\mu_1} \right\rfloor, m_2 = \left\lfloor \frac{\lambda}{1-\mu_2} \right\rfloor$ and k defined in (4.4) is constant. We also take constant values of a and b in I, so that g, h, t, s, r are also fixed.

As in [8], we define for $\lambda \in I$ the quantities

$$\begin{aligned} H &= Z_0 + Z_1\lambda, \\ Z_0 &= \frac{\left(m_1^2 + m_1\right)(1-\mu_1) + \left(m_2^2 + m_2\right)(1-\mu_2) - h^2 + h - (1-\mu_1-\mu_2)\left(g^2+g\right)}{2}, \\ Z_1 &= h + g - m_1 - m_2 - 1 = a - b \in \{-1, 0, 1\}, \end{aligned}$$

so that

$$H \geq H' := Z_0 + \begin{cases} \lambda_1 & Z_1 = 1 \\ 0 & Z_1 = 0 \\ -\lambda_2 & Z_1 = -1 \end{cases}.$$

It follows that

$$E \leq \frac{\log\left(Y\lambda_1^2\right)}{Y\lambda_1^4} - \frac{H' - 0.001\mu_1k^2 - \mu_2\left(\frac{t(t-1)}{2} + \frac{s^2}{\xi t g^{3/2}} + hte^{-s/(ht)}\right)}{2rs} := E'.$$

At this point we use Program 3 in [2] to find the best values for C and u, under the condition $u \leq 132.94357$, so that, for $\lambda \in I$,

$$S(N,t) \leq CN^{1-1/(u\lambda^2)} + \frac{1}{k}N^{1-1/132\lambda^2},$$

where $u = 1/\left(E'\lambda_1^2\right)$ and $C = C_3^{1/r}\left(\lambda^{4.65\lambda}C_1C_2\right)^{1/(2rs)}$. Running Program 3 we can notice that in each interval I we have $C \leq 8.7633 \leq 8.7979$ and $u \leq 132.94357$. The conclusion follows. □

Lemma 4.2. *For $\lambda \geq 220$ and $N \geq e^{288\lambda^2}$ we have the following estimate*

$$S(N,t) \leq 7.5N^{1-1/132.94357\lambda^2}.$$

Proof. From Lemma 2.9, (4.2), (4.4), (4.6), (4.7), (4.8) and (4.9) we have

$$\begin{aligned} S(N,t) &\le \left(C_3^{\frac{1}{r}}\left(\lambda^{4.65\lambda}C_1C_2\right)^{\frac{1}{2rs}}\right)N^{1+E} + 2N^{0.36} + \frac{1}{k}N^{1-0.0000019476} \\ E &= \frac{\log\left(Y\lambda^2\right)}{7.6Y\lambda^4} + \frac{1}{2rs}\left(-H + 0.001\mu_1k^2 + \mu_2E_2\right). \end{aligned} \tag{4.11}$$

Now, we assume

$$h = \left\lfloor 1.17928\lambda + \frac{1}{2}\right\rfloor, \quad g = \left\lfloor 1.24788\lambda + \frac{1}{2}\right\rfloor, \quad s = \lfloor \sigma h(t-1) + 1\rfloor, \tag{4.12}$$

where

$$\sigma = 0.3299, \quad t = g - h + 1. \tag{4.13}$$

From (4.1), (4.12) and (4.13), the relation (4.5) holds and, furthermore, we have

$$|\gamma - 1.17928| \le \frac{1}{2\lambda}, \quad |\phi - 1.24788| \le \frac{1}{2\lambda}.$$

From (4.4), (4.12) and (4.13) we have

$$g \ge 275, \quad h \ge 259, \quad t \ge 17 \ge 13, \quad k \ge 338, \quad s \ge 0.02668\lambda^2 \ge 0.02294\lambda^2.$$

Following exactly the proof of Lemma 5.2 in [8], with $Y = 288$ instead of $Y = 300$, with the adjusted value of $D = 0.1090Y = 29.3472$ instead of $D = 30.57$, we still have

$$C_3^{1/r}\left(\lambda^{4.65\lambda}C_1C_2\right)^{1/2rs} \le 7.5.$$

It remains to estimate E. We have

$$\begin{aligned} E \le& \frac{\log\left(Y\lambda^2\right)}{7.6Y\lambda^4} + \frac{-H + 0.001\mu_1k^2}{2.00002\rho\sigma\gamma(\phi-\gamma)\lambda^2k^2} + \frac{\mu_2E_2}{2\rho k^2 s} \\ \le& \frac{1.52\times10^{-7}}{\lambda^2} + \frac{-\lambda^2H_2 - \lambda H_1 + H_0}{2.00002\rho\sigma\gamma(\phi-\gamma)\lambda^2k^2} + \frac{0.001\mu_1}{2.00002\rho\sigma\gamma(\phi-\gamma)\lambda^2} \\ &+ \frac{\mu_2}{2\rho k^2}\left[\frac{\phi-\gamma+1/\lambda}{2\sigma\gamma} + \frac{(t/(t-1))e^{-\sigma+\sigma/t}}{\sigma} + \frac{\sigma h g^{-3/2}}{12}\right]. \end{aligned}$$

Following Ford's argument in [8], we have

$$\lambda^2E \le 1.52\times10^{-7} + \frac{f(\gamma,\phi) + G_1/\lambda^{1/2} + G_2/\lambda}{\rho}$$

where

$$\begin{aligned} f(\gamma,\phi) &= \frac{1}{2.00002\sigma\gamma}\left[\frac{0.001\mu_1}{\phi-\gamma} + \frac{1}{k_1^2}\left(\frac{-H_2}{\phi-\gamma} + 1.00001\mu_2\left(\frac{1}{2}(\phi-\gamma) + \gamma e^{-\sigma}\right)\right)\right], \\ G_1 &= \frac{\mu_2\sigma\gamma\phi^{-3/2}}{24k_0^2} \le 0.0008, \\ G_2 &= \frac{1}{2.00002\sigma(k/\lambda)^2}\left[\frac{-H_1 + H_0/\lambda + 1.33547\mu_2\gamma e^{-\sigma}}{\gamma(\phi-\gamma)} + \frac{1.00001\mu_2}{2\gamma}\right]. \end{aligned}$$

Using Ford's estimate for the expression inside the brackets in the definition of G_2, we have

$$G_2 \leq \frac{0.0392}{2.00002\sigma\gamma k_0^2} \leq 0.0213552.$$

It follows that

$$\begin{aligned}\lambda^2 E &\leq 1.56 \times 10^{-7} + \frac{f(\gamma,\phi) + 0.0008\lambda^{-1/2} + 0.0213552\lambda^{-1}}{\rho} \\ &\leq 0.0000473 + \frac{f(\gamma,\phi)}{\rho}.\end{aligned}$$

In the range

$$|\phi - 1.24788| \leq \frac{1}{440}, \qquad |\gamma - 1.17928| \leq \frac{1}{440},$$

$f(\gamma,\phi)$ is increasing, hence the maximum occurring at $\gamma = 1.17928 + \frac{1}{440}, \phi = 1.24788 - \frac{1}{440}$, with

$$f(\gamma,\phi) \leq -0.024287046496.$$

Using $\rho = 3.208\ 63$ from Theorem 1.4, it follows that

$$\lambda^2 E \leq -\frac{1}{132.94357}. \quad \square$$

Theorem 1.5 for $\lambda \geq 84$ follows directly from Lemma 4.1 and Lemma 4.2.

Remark. One might have considered the ranges $84 \leq \lambda \leq N$ and $\lambda > N$, with $N > 220$. However, a direct computation shows that even for really large values of N, of the order of 10^6, and hence, by definition of k, for values of k much greater than 90 000, the improvement is negligible. Indeed, one can see that a new choice for N would not influence the estimate for $S(N,t)$ in the range $84 \leq \lambda \leq N$ we found in Lemma 4.1. For the case $\lambda > N$, if N (and hence k) is sufficiently large then the value of ρ in Theorem 1.4 would be 3.208 61 which is slightly smaller than $\rho = 3.208\ 63$ which we found for $k \geq 90\ 000$. However, a direct computation shows that this really small improvement on ρ for k sufficiently large would not lead to an improvement in Lemma 4.2, when $\lambda > N$ and N sufficiently large.

4.2. Case $\lambda \leq 84$

For $\lambda \leq 84$, the values for the C constant found in [8] hold, both for $1 \leq \lambda \leq 2.6$ and $2.6 \leq \lambda \leq 84$. Indeed, for the range $2.6 \leq \lambda \leq 84$, the same values for C still hold under the new constraint that $S(N,t) \leq CN^{1-1/u\lambda^2}$ with $u \leq 132.94357$. We can notice that the maximum is reached inside the interval $\lambda \in [83, 84]$, where $C = 8.7979$. This concludes the proof of Theorem 1.5 also for $\lambda \leq 84$.

5. Proof of Theorem 1.1

First of all, since $\zeta(\bar{s},u) = \overline{\zeta(s,u)}$ and $\zeta(s) = \zeta(s,1)$, we restrict our attention to s lying in the upper half-plane. Then, we consider separately the cases $\sigma \leq \frac{15}{16}$ or $t \leq 10^{108}$ or $t \geq 10^{108}$ and $\frac{15}{16} \geq \sigma \geq 1$. The main contribution will come from the case $t \geq 10^{108}$ and $\frac{15}{16} \geq \sigma \geq 1$.

5.1. Case $\frac{15}{16} \leq \sigma \leq 1,\ t \geq 10^{108}$

Using Lemma 2.12 with the values $C = 8.7979$ and $D = 132.94357$ found in Theorem 1.5 and $t \geq 10^{108}$, we have

$$\left(\frac{C+1+10^{-80}}{\log^{2/3} t} + 1.569CD^{1/3}\right) \leq \left(\frac{C+1+10^{-80}}{\log^{2/3}(10^{108})} + 1.569CD^{1/3}\right) \leq 70.6995.$$

It follows that

$$|\zeta(\sigma+it)| \leq 70.6995 t^{4.43795(1-\sigma)^{3/2}} \log^{2/3} t$$

and

$$\left|\zeta(s,u) - u^{-s}\right| \leq 70.6995 t^{4.43795(1-\sigma)^{3/2}} \log^{2/3} t.$$

5.2. Case $\frac{15}{16} \leq \sigma \leq 1,\ 3 \leq t \leq 10^{108}$

Following the proof of Lemma 2.10 in [8], one has

$$\left|\zeta(s,u) - u^{-s}\right| \leq (t+3/2)^{1-\sigma}(1 + 1/t + \log(2t+1))$$

for $\frac{15}{16} \leq \sigma \leq 1$ and $3 \leq t \leq 10^{108}$.

If $3 \leq t \leq 10^6$, from [8] we know that

$$\left|\zeta(s,u) - u^{-s}\right| \leq 36.8.$$

If $10^6 \leq t \leq 10^{108}$, from the argument of Lemma 2.10 in [8], one gets

$$\begin{aligned}
\left|\zeta(s,u) - u^{-s}\right| &\leq (t+3/2)^{1-\sigma}(1 + 1/t + \log(2t+1)) \\
&\leq 1.123 t^{1-\sigma} \log t \\
&= 1.123 \left(t^{4(1-\sigma)^{3/2}} \log^{2/3} t\right) \left(t^{1-\sigma-4(1-\sigma)^{3/2}} \log^{1/3} t\right) \\
&\leq 1.123 \left(t^{4(1-\sigma)^{3/2}} \log^{2/3} t\right) \left(t^{\frac{1}{108}} \log^{1/3} t\right) \\
&\leq 70.6199 t^{4(1-\sigma)^{3/2}} \log^{2/3} t \\
&\leq 70.6995 t^{4.43795(1-\sigma)^{3/2}} \log^{2/3} t.
\end{aligned}$$

Hence, we also have

$$|\zeta(\sigma+it)| \leq 70.6199 t^{4(1-\sigma)^{3/2}} \log^{2/3} t \leq 70.6995 t^{4.43795(1-\sigma)^{3/2}} \log^{2/3} t.$$

Remark. We note that the estimate found for $\zeta(s)$ and $\zeta(s,u)$ in this range is much better than the one found for $\frac{15}{16} \leq \sigma \leq 1$ and $t \geq 10^{108}$. Indeed, the constant B we found in this current case is just 4, which is less than the final value 4.43795.

5.3. Case $\frac{1}{2} \le \sigma \le \frac{15}{16},\ t \ge 3$

From the proof of Lemma 2.10 in [8] we have

$$|\zeta(\sigma + it)| \le 21.3t^{4(1-\sigma)^{3/2}} \le 70.6995t^{4.43795(1-\sigma)^{3/2}} \log^{2/3} t. \tag{5.1}$$

Remark. The choice of $\frac{15}{16}$ comes from the proof of Lemma 7.1 in [8], where $(1-\sigma) \le 4(1-\sigma)^{3/2}$ for $\sigma \le \frac{15}{16}$.

With $(1-\sigma) \le 4.4(1-\sigma)^{3/2}$ one should get $\sigma \le \frac{459}{484}$. However, the influence of this second choice on the A constant is negligible and it does not lead to any further improvements on A.

6. Proof of Theorem 1.2

For $t \le \exp(481958)$ we use the best known explicit Littlewood zero-free region due to Yang in [32]. For $t \ge \exp(481958)$ we follow the argument used to prove Theorem 1.1 in [18] which relies on a non-negative trigonometric polynomial $P_{40}(x)$ defined by (1.6) with degree 40 having

$$b_0 = 1, \quad b_1 = 1.746\ 001\ 909\ 149\ 94, \quad b = \sum_{k=1}^{40} b_k = 3.564\ 539\ 654\ 371\ 34.$$

Using the new values $A = 70.6995$ and $B = 4.43795$ found in Theorem 1.1 and making the following new assumptions

$$T_0 := \exp(481958), \qquad M_1 := 0.050\ 021, \qquad E = 1.796\ 6797, \qquad R = 469$$

in [18], the proof of Theorem 1.2 is complete.

7. Proof of Theorem 1.3

The proof is exactly the same as in [18], except for the new value $B = 4.43795$ found in Theorem 1.1, and relies on a non-negative polynomial $P_{46}(x)$ with degree 46 where

$$b_0 = 1, \quad b_1 = 1.747\ 087\ 440\ 818\ 48, \quad b = \sum_{k=1}^{46} b_k = 3.574\ 409\ 430\ 220\ 73.$$

8. Computation

Using double precision arithmetic, it is a relatively simple matter to implement the required code. Program 1 takes a range for k and computes allowable values for ρ and θ. Program 2 is very similar but works for $500 \le k < 90000$, in which case we can assume $\Delta_n \le 0.4k^2$ and $n_0 = \lceil 0.1247k \rceil$. We then run Program 2 at the end points $k = 500$ and $k = 89\,999$. The final double precision program, Program 3, has the outputs from Programs 1 and 2 hard coded into it and finds the best values for C and $u \le 132.94357$ for $\lambda \in [84, 220]$.

At the suggestion of the referee, we implemented the same programs using interval arithmetic to ensure that our results were rigorous. We combined Programs 1 and 2 and coded that and Program 3 using Platt's double precision interval library (see §4 of [19]). We found that the interval version of Programs 1 and 2 lost too much precision and so we turned to used Johansson's ARB [11] running at 100 bits of precision. Double precision was ample for Program 3.

The Programs used are available on github at [2].

9. Some possible improvements on A

In this section we give some suggestions for some possible improvements on the constant A in Theorem 1.1. We will provide some quite detailed proofs of some useful lemmas to help the reader follow the argument more easily.

Instead of the definition (1.7) for $S(N,t)$, one could consider the following sum

$$\tilde{S}(N,t) := \max_{0<u\le 1} \max_{N<R\le mN} \left| \sum_{N\le n\le R} \frac{1}{(n+u)^{it}} \right| \tag{9.1}$$

with $1 < m \le 2$. For $\lambda \ge 84$, we can find sharper estimates.

Theorem 9.1. *For* $1.001 \le m \le 2$ *and* $\lambda \ge 84$ *we have:*

$$\tilde{S}(N,t) \le (m-1)8.7979N^{1-1/132.94357\lambda^2}.$$

If one could find a bound of the form $\tilde{S}(N,t) \le (m-1)cN^{1-1/(u\lambda^2)}$ also for $\lambda \le 84$, then the constant A in Theorem 1.1 might be reduced to around 49. We briefly outline the proof of Theorem 9.1.

We start with a preliminary lemma that is a more general version of Lemma 2.9.

Lemma 9.2. *Suppose* k, r *and* s *are integers* ≥ 2*, and* h *and* g *are integers satisfying* $1 \le h \le g \le k$*. Let* N *be a positive integer, and* M_1, M_2 *be real numbers with* $1 \le M_i \le N$*. Let* $\mathscr{B}$ *be a nonempty subset of the positive integers* $\le M_2$*. Then*

$$\tilde{S}(N,t) \le 2M_1M_2 + \frac{(m-1)t(M_1M_2)^{k+1}}{(k+1)N^k}$$
$$+ (m-1)N\left(\frac{M_2}{|\mathscr{B}|}\right)^{1/r} \left((5r)^k M_2^{-2s} \lfloor M_1 \rfloor^{-2r+k(k+1)/2} J_{r,k}(\lfloor M_1 \rfloor) J_{s,g,h}(\mathscr{B}) W_h \dots W_g\right)^{1/2rs},$$

where

$$W_j = \min\left(2sM_2^j, \frac{2sM_2^j}{r\lfloor M_1 \rfloor^j} + \frac{stM_2^j}{\pi j N^j} + \frac{4\pi j(2N)^j}{rt\lfloor M_1 \rfloor^j} + 2\right) \quad (j \ge 1)$$

and $\tilde{S}(N,t)$ *is defined in* (9.1)*.*

Proof. We define $M = \lfloor M_1 \rfloor$. For $N < R \le mN$, $1 < m \le 2$ and $0 < u \le 1$, we have

$$\left| \sum_{N<n\le R} (n+u)^{-it} \right| = \frac{1}{M|\mathscr{B}|} \left| \sum_{\substack{a\le M_1 \\ b\in\mathscr{B}}} \sum_{N<n+ab\le R} (n+ab+u)^{-it} \right|$$
$$\le \frac{1}{M|\mathscr{B}|} \left| \sum_{\substack{a\le M_1 \\ b\in\mathscr{B}}} \sum_{N<n\le R-1} (n+ab+u)^{-it} \right| + \frac{1}{M|\mathscr{B}|} \sum_{\substack{a\le M_1 \\ b\in\mathscr{B}}} (2ab-1)$$
$$\le \frac{N}{M|\mathscr{B}|} \max_{N\le z\le mN} \left| \sum_{\substack{a\le M_1 \\ b\in\mathscr{B}}} e^{-it\log(1+ab/z)} \right| + 2M_1M_2.$$

For $0 \leq x \leq 1$ we have

$$\left|\log(1+x) - \left(x - x^2/2 + \cdots + (-1)^{k-1}x^k/k\right)\right| \leq \frac{x^{k+1}}{k+1}.$$

Also $\left|e^{iy} - 1\right| \leq y$ for real y and $ab/z \leq M_1M_2/N$. Thus, for some $z \in [N, mN]$,

$$\tilde{S}(N,t) \leq \frac{(m-1)N}{M|\mathscr{B}|}|U| + \frac{t(m-1)\left(M_1M_2\right)^{k+1}}{(k+1)N^k} + 2M_1M_2$$

where $U = \sum_{a,b} e\left(\gamma_1(ab) + \cdots + \gamma_k(ab)^k\right)$ and $\gamma_j = (-1)^j t/\left(2\pi j z^j\right)$.

At this point, from Ford's argument in Lemma 5.1 of [8], we have

$$\frac{N}{M|\mathscr{B}|}|U| \leq N\left(\frac{M_2}{|\mathscr{B}|}\right)^{\frac{1}{r}}\left((5r)^k M_2^{-2s}\left\lfloor M_1\right\rfloor^{-2r+\frac{1}{2}k(k+1)} J_{r,k}\left(\left\lfloor M_1\right\rfloor\right) J_{s,g,h}(\mathscr{B}) W_h \cdots W_g\right)^{\frac{1}{2rs}}.$$

The conclusion follows. □

At this point, from Ford's argument in section 5 of [8] we have the following bounds:

$$\frac{t(M_1M_2)^{k+1}}{(k+1)N^k} \leq \frac{1}{k}N^{1-0.0000019476},$$
$$2M_1M_2 \leq 2N^{0.36}$$

and

$$N\left(\frac{M_2}{|\mathscr{B}|}\right)^{1/r}\left((5r)^k M_2^{-2s}\lfloor M_1\rfloor^{-2r+k(k+1)/2} J_{r,k}(\lfloor M_1\rfloor) J_{s,g,h}(\mathscr{B}) W_h \ldots W_g\right)^{1/2rs}$$
$$\leq \left(C_3^{\frac{1}{r}}\left(\lambda^{5\lambda} C_1C_2\right)^{\frac{1}{2rs}}\right) N^{1+E}$$

where

$$E = \frac{\log\left(Y\lambda^2\right)}{7.6Y\lambda^4} + \frac{1}{2rs}\left(-H + 0.001\mu_1 k^2 + \mu_2 E_2\right)$$

and C_1, C_2, C_3 are the same as for Theorem 1.5. If we use these bounds in Lemma 9.2 we get

$$\tilde{S}(N,t) \leq (m-1)\left(C_3^{\frac{1}{r}}\left(\lambda^{4.65\lambda} C_1C_2\right)^{\frac{1}{2rs}}\right) N^{1+E} + 2N^{0.36} + \frac{(m-1)}{k}N^{1-0.0000019476}. \tag{9.2}$$

Now, we consider $N \geq e^{288\lambda^2}$, since otherwise we have trivially

$$\tilde{S}(N,t) \leq (m-1)N \leq (m-1)e^{288/132.94357}N^{1-1/132.94357\lambda^2} \leq (m-1)8.7979N^{1-1/132.94357\lambda^2}.$$

Lemma 9.3. *For $84 \leq \lambda \leq 220$ and $N \geq e^{288\lambda^2}$ we have the following estimate:*

$$\tilde{S}(N,t) \leq (m-1)8.797901N^{1-1/132.94357\lambda^2}.$$

Proof. From Lemma 4.1 we know that

$$\left(C_3^{\frac{1}{r}}\left(\lambda^{4.65\lambda} C_1C_2\right)^{\frac{1}{2rs}}\right) N^{1+E} \leq 8.7979N^{1-1/132.94357\lambda^2}.$$

Using this result in (9.2) we get

$$\begin{aligned}\tilde{S}(N,t) &\leq (m-1)8.7979N^{1-1/132.94357\lambda^2} + 2N^{0.36} + \frac{(m-1)}{k}N^{1-0.0000019476} \\ &= (m-1)\left(8.7979N^{1-1/(132.94357\lambda^2)} + \frac{2N^{0.36}}{m-1} + \frac{1}{k+1}N^{1-1/132\lambda^2}\right) \\ &\leq (m-1)8.797901N^{1-1/(132.94357\lambda^2)}\end{aligned}$$

where the last passage comes from the fact that, for $m = 1.001$, we have

$$(m-1)\cdot 10^{-7}\cdot 8.7979N^{1-1/(132.94357\lambda^2)} \geq 2N^{0.36}.$$

Indeed, for $84 \leq \lambda \leq 220$ and $N \geq e^{288\lambda^2}$, we have

$$\begin{aligned}N^{1-1/(132.94357\lambda^2)-0.36} &\geq e^{288\lambda^2(1-1/(132.94357\lambda^2)-0.36)} \\ &\geq e^{288\cdot 84^2(1-1/(132.94357\cdot 84^2)-0.36)} \\ &\geq \frac{2}{8.7979\cdot 10^{-7}\cdot(m-1)}\end{aligned}$$

if

$$m-1 \geq \frac{2\cdot 10^7}{8.7979}\cdot e^{-288\cdot 84^2(1-1/(132.94357\cdot 84^2)-0.36)} \geq 10^{-560000},$$

which holds, since with our choice $m-1 = 10^{-3}$. □

Lemma 9.4. *For $\lambda \geq 220$ and $N \geq e^{288\lambda^2}$ we have the following estimate:*

$$\tilde{S}(N,t) \leq (m-1)7.5001N^{1-1/132.94357\lambda^2}.$$

Proof. From Lemma 4.2 we know that

$$\left(C_3^{\frac{1}{r}}\left(\lambda^{4.65\lambda}C_1C_2\right)^{\frac{1}{2rs}}\right)N^{1+E} \leq 7.5N^{1-1/132.94357\lambda^2}.$$

Using this result in (9.2) we get

$$\begin{aligned}\tilde{S}(N,t) &\leq (m-1)7.5N^{1-1/132.94357\lambda^2} + 2N^{0.36} + \frac{(m-1)}{k}N^{1-0.0000019476} \\ &\leq (m-1)\left(7.5N^{1-1/(132.94357\lambda^2)} + \frac{2N^{0.36}}{m-1} + \frac{1}{k+1}N^{1-1/132.94357\lambda^2}\right) \\ &\leq (m-1)7.5001N^{1-1/(132.94357\lambda^2)}\end{aligned}$$

where the last passage comes from the fact that, for $m = 1.001$, we have

$$(m-1)\cdot 10^{-5}\cdot 7.5N^{1-1/(132.94357\lambda^2)} \geq 2N^{0.36}.$$

Indeed, for $\lambda \geq 220$ and $N \geq e^{288\lambda^2}$, we have

$$\begin{aligned} N^{1-1/(132.94357\lambda^2)-0.36} &\geq e^{288\lambda^2(1-1/(132.94357\lambda^2)-0.36)} \\ &\geq e^{288\cdot 220^2(1-1/(132.94357\cdot 220^2)-0.36)} \\ &\geq \frac{2}{7.5\cdot 10^{-5}(m-1)} \end{aligned}$$

if

$$m-1 \geq \frac{2\cdot 10^5}{7.5}\cdot e^{-288\cdot 220^2(1-1/(132.94357\cdot 220^2)-0.36)} \geq 10^{-3800000},$$

which holds, since with our choice $m-1 = 10^{-3}$. □

Theorem 9.1 follows immediately from Lemma 9.3 and Lemma 9.4.

At this point, if one could find an estimate of the form $\tilde{S}(N,t) \leq (m-1)cN^{1-1/(u\lambda^2)}$ also for $\lambda \leq 84$, then we can use a more general version of Lemma 2.12.

Lemma 9.5. *If $s = \sigma + it$, $\frac{15}{16} \leq \sigma \leq 1, t \geq 10^{90}$ and $0 < u \leq 1$, then*

$$\left|\zeta(s,u) - \sum_{0\leq n\leq t}(n+u)^{-s}\right| \leq 10^{-77}.$$

Proof. The proof is the same as that one of Lemma 2.11 in [8], with $t \geq 10^{90}$ instead of $t \geq 10^{100}$. □

Lemma 9.6. *Suppose that $\tilde{S}(N,t) \leq (m-1)CN^{1-1/(D\lambda^2)} (1 \leq N \leq t)$ for positive constants C and D and $1 < m \leq 2$, where $\lambda = \frac{\log t}{\log N}$. Let $B = \frac{2}{9}\sqrt{3D}$. Then, for $\frac{15}{16} \leq \sigma \leq 1, t \geq 10^{90}$ and $0 < u \leq 1$, we have*

$$|\zeta(s)| \leq \left(\frac{(m-1)C + 1 + \frac{1}{10^{77}}}{\log^{2/3} t} + \frac{1.0875034(m-1)CD^{1/3}}{\log(m)}\right) t^{B(1-\sigma)^{3/2}} \log^{2/3} t.$$

Proof. Let

$$S_1(u) = \sum_{1\leq n\leq t}(n+u)^{-s}$$

By Lemma 9.5, $|\zeta(s,u) - u^{-s}| \leq 10^{-77} + S_1(u)$. Put $r = \left\lceil \frac{\log t}{\log m} \right\rceil$. By partial summation,

$$\begin{aligned} |S_1(u)| &\leq 1 + \sum_{j=0}^{r-1}\left|\sum_{m^j < n \leq \min(t, m^{j+1})}(n+u)^{-\sigma-it}\right| \\ &\leq 1 + \sum_{j=0}^{r-1}\left(m^j\right)^{-\sigma} S\left(m^j, t\right) \\ &\leq 1 + (m-1)C\sum_{j=0}^{r-1} e^{g(j)}, \end{aligned}$$

where

$$g(j) = (1-\sigma)(j\log m) - \frac{(j\log m)^3}{D\log^2 t}.$$

As a function of x, $g(x)$ is increasing on $[0, x_0]$ and decreasing on $[x_0, \infty)$, where $x_0 \log m = \sqrt{D(1-\sigma)/3}\log t$. Thus

$$\begin{aligned}\frac{|S_1(u)|-1}{(m-1)C} &\le e^{g(x_0)} + \int_0^r e^{g(x)}dx \\ &\le t^{B(1-\sigma)^{3/2}} + \frac{D^{1/3}\log^{2/3} t}{\log m}\int_0^\infty e^{3y^2u-u^3}du,\end{aligned}$$

where $y = \sqrt{(1-\sigma)/3}D^{1/6}\log^{1/3} t$.

To bound the last integral, we make use of the inequality

$$e^{-2y^3}\int_0^\infty e^{3y^2u-u^3}du \le 1.0875034 \quad (y \ge 0),$$

where the maximum occurs near $y = 0.710$. Therefore

$$\frac{|S_1(u)|-1}{(m-1)C} \le t^{B(1-\sigma)^{3/2}}\left(1 + \frac{1.0875034}{\log m}D^{1/3}\log^{2/3} t\right) \quad \square$$

However, Ford's method in [8] for $\lambda \le 87$ (or $\lambda \le 84$ as in our paper), cannot be modified to extract the factor $(m-1)$. Indeed, following the proof of Lemma 6.3 in [8], one should estimate the following quantity

$$\left(\sum_{N<n\le mN-1}|T(n)|^{2s}\right)^{1/2s},$$

where

$$T(n) = \sum_{v\le M} e\left(-\frac{t}{2\pi}\log\left(1+\frac{v}{n+u}\right)\right).$$

However, since an estimate on the whole sum of $T(n)$ over $N \le n \le mN$ is required, instead of one for the maximum of $T(n)$ over the interval $N \le n \le mN$, it is not possible to extract a factor $(m-1)$ at this step. A clever argument which would overcome this problem could lead to a suitable estimate also for the case $\lambda \le 84$, with a consequent improved value for A.

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